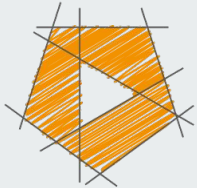




Level 1 Enterprise Architect

Training for the Level 1 exam for the TOGAF® Standard, 10th Edition



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TOGAF® Standard, 10th Edition

**Certification disconnected
from TOGAF standard version**

**New version introduced in
2022**

**Very little change in the
fundamental content from
version 9.X to version 10.0**

—



What This Course Covers

Preparation for the Level 1 syllabus for Enterprise Architecture Foundation

Concepts & Definitions

The ADM & ADM Techniques

Applying the ADM

Architecture Governance

Architecture Content

**Industry leading standard for
enterprise architecture**

**Supported by hundreds of
organizations worldwide**

**I can't promise “fun”, but I will
try like heck to avoid “boring”**

Ready?

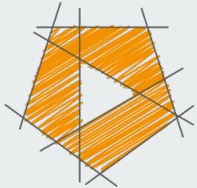
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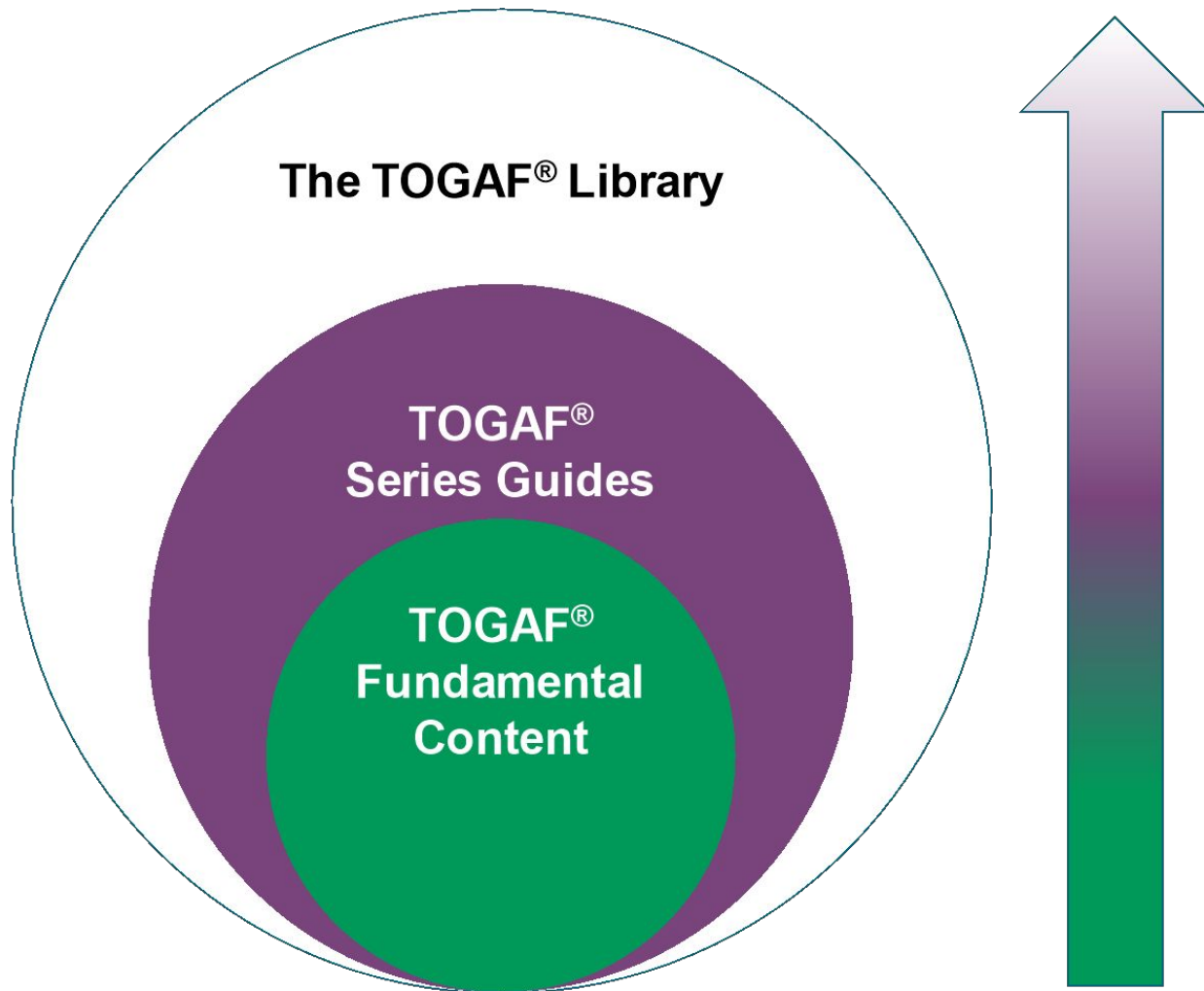
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Changes from version 9.X to version 10.0

**Content has been divided from
one big spec to 18 separate
documents**

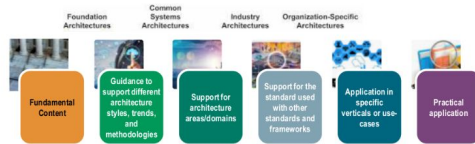


The TOGAF Documentation Set

The Open Group Library

Enterprise Architecture-Related Content

The TOGAF® Library



The TOGAF® Standard

TOGAF® Fundamental Content

- Introduction and Core Concepts
- Architecture Development Method
- ADM Techniques
- Applying the ADM
- Architecture Content
- Enterprise Architecture Capability and Governance

TOGAF® Series Guides



POCKET
GUIDES



WHITE
PAPERS



REFERENCE
ARCHITECTURES



DATA
SHEETS



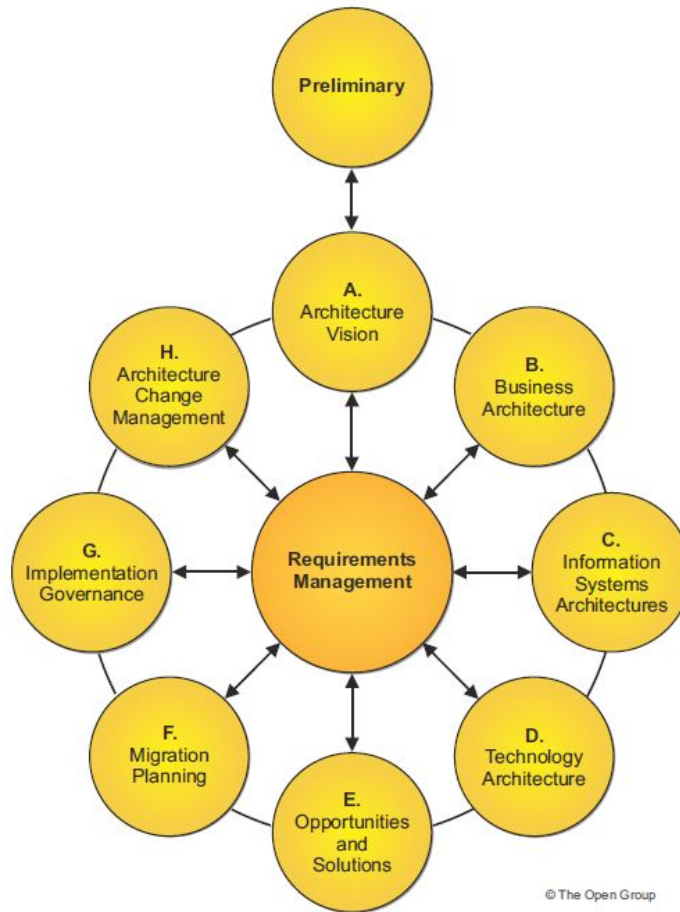
REFERENCE
CARDS



GUIDES

**Removal of arrows between
the phases to avoid a waterfall
model**

—



© The Open Group

Figure 1-1 Architecture Development Cycle

The term *Content Framework and Metamodel* is changed to *TOGAF Content Framework and TOGAF Enterprise Metamodel*

**The term “Standards
Information Base” is replaced
by “Standards Library”.**

The term “Governance Log” is replaced by “Governance Repository”.

—

**The version numbers
associated with deliverables
are replaced: 0.1 by draft, 1.0
by approved**

—

**The concept of Architecture
Alternatives is new**

**Architecture Building Blocks
now abbreviated as ABBs**

**Solution Building Blocks now
abbreviated as SBBs**

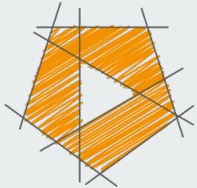
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Trying not to talk about myself

Try harder, Scott

—

**In the IT industry for over 20
years**

**Teaching TOGAF online for 9
years**

—

**Over 180,000 students in the
TOGAF 9.1 and 9.2 courses**

—

Member of The Open Group



**Licensed to teach TOGAF
commercially**

**Goal of the course is to pass
the exam**

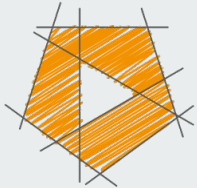
**And help you understand this
18-document standard in an
easier way**





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What is TOGAF, Anyways?

An architecture framework for enterprise architecture

What's a framework?

Methods, tools, and common definitions

Architecture Development Method (ADM)

What is architecture?

“It’s complicated”

**ISO/IEC/IEEE
42010 : 2011**

—



ISO/IEC/IEEE 42010: 2011

“The fundamental concepts or properties of a system in its environment embodied in its elements, relationships, and in the principles of its design and evolution.”

TOGAF 10 agrees with that. But also adds.



TOGAF 10 Definition of Architecture.

“The structure of components, their inter-relationships, and the principles and guidelines governing their design and evolution over time.”

**Both are saying the same
thing.**

**An architecture is the
components of a system and
their relationship.**

**AND the principles governing
their design and evolution over
time.**

—

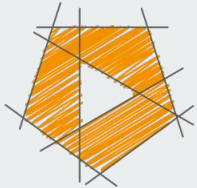
**An architecture framework is
how we do architecture.**





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TOGAF Certification for People Program

Two Levels

Level 1 and Level 2

**Passing the Level 1 gives you
TOGAF® Enterprise Architecture Foundation**

**Passing Level 2 gives you
TOGAF® Enterprise Architecture Practitioner**



TOGAF Level 1

- A test of core knowledge of the standard
- A multiple-choice test, only 1 correct answer
- 40 questions over 60 minutes
- Need 55% to pass, or 22 correct out of 40
- Each correct answer is worth 1 point



TOGAF Level 2

- A test of how you apply the TOGAF Standard
- A scenario based test, scored using gradient scoring
- Open book exam
- 8 questions over 90 minutes
- Need 60% to pass
- Must have passed Level 1 to take Part 2

**US\$395 per test
(as of Aug 1, 2023)**

—

**US\$595 for the combined test
(as of Aug 1, 2023)**

—

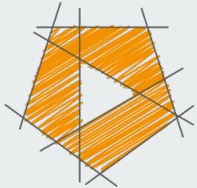
English





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TOGAF 10 Standard Document

**Recommend you read the
relevant chapter with each
lesson**

—

**Free HTML or PDF version on
The Open Group website**

The TOGAF Standard: Introduction and Core Concepts

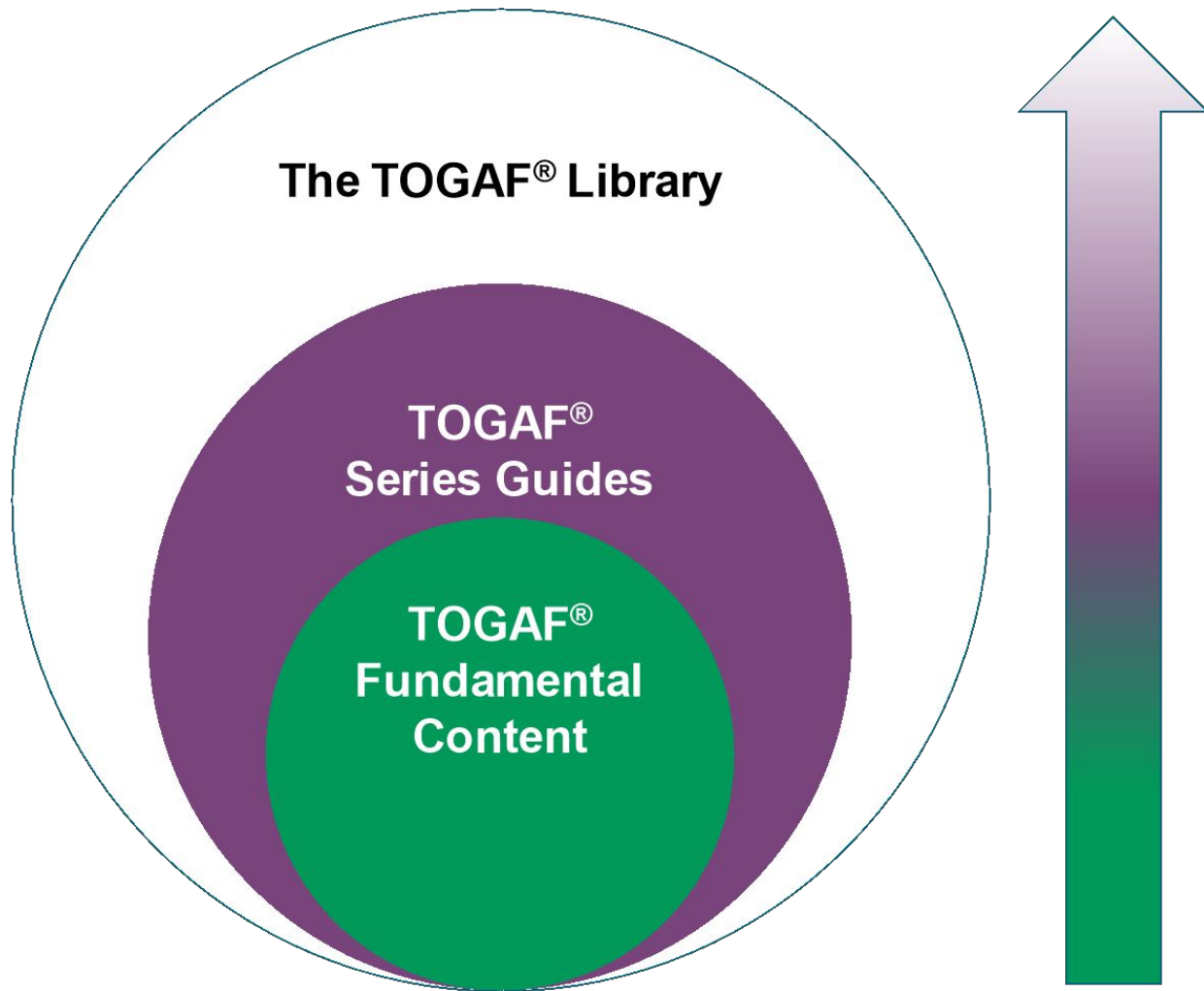
The TOGAF Standard: Architecture Development Method (ADM)

The TOGAF Standard: ADM Techniques

The TOGAF Standard: Applying the ADM

The TOGAF Standard: Architecture Content

The TOGAF Standard: Enterprise Architecture Capability and Governance



The TOGAF® Standard

TOGAF® Fundamental Content

- Introduction and Core Concepts
- Architecture Development Method
- ADM Techniques
- Applying the ADM
- Architecture Content
- Enterprise Architecture Capability and Governance

TOGAF® Series Guides

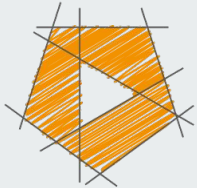
- Business Architecture
- Information Architecture
- Security Architecture
- Enterprise Architecture/Agile Architecture
- Enterprise Architecture/Digital Enterprise
- Technology Architecture
- MSA/SOA Architectures
- Adapting the ADM
- ...





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**“The core concepts of the
TOGAF 10 Standard”**

—



7 Core Concepts

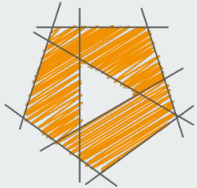
- The definition of enterprise
- The architecture domains, or BDAT
- The architecture development method, or ADM
- Deliverables, artifacts and building blocks
- The enterprise continuum
- The architecture repository
- And, defining architecture capability





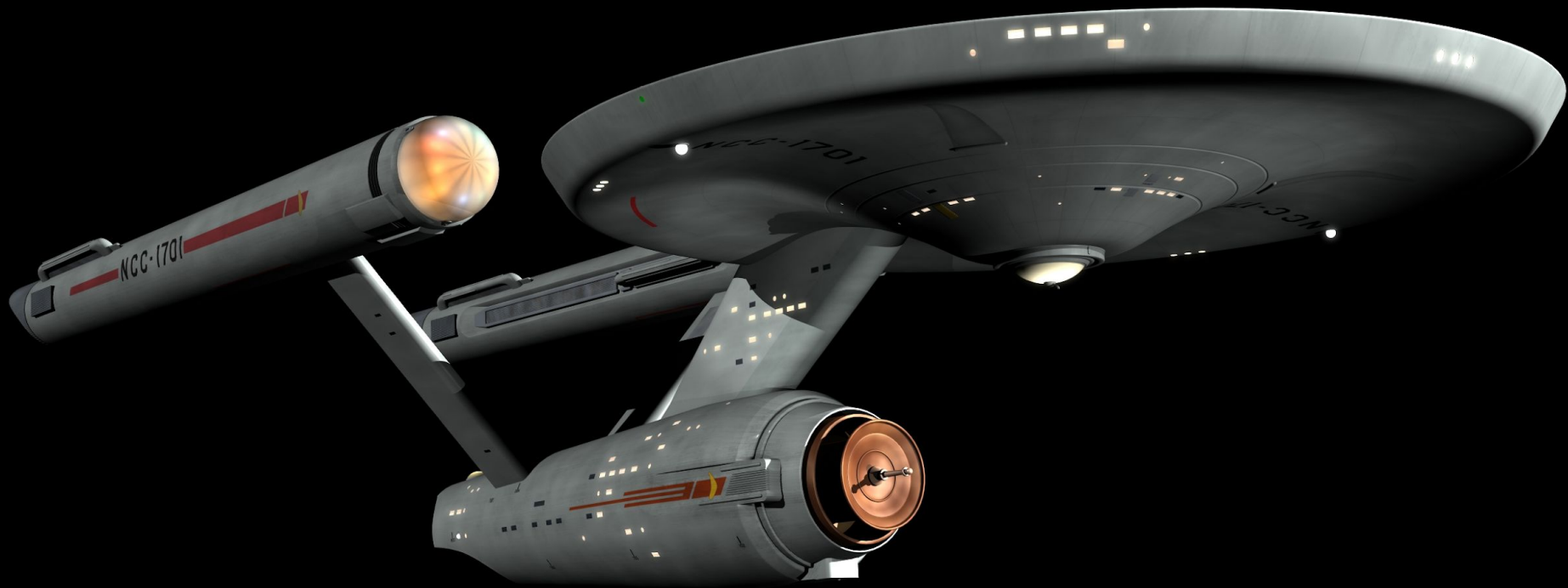
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The definition of Enterprise



Not that one.

**“Highest level of description
for an organization”**

**Board of directors, executive
team, the deciders**

—

**Could be a collection of
organizations.**

**“Highest level of description
for an organization”**

Scope of enterprise architecture

Maybe you are an enterprise architect but not responsible for the whole enterprise.

**The enterprise might be
scoped differently for EA.**

**You should know where that
scope is before you start.**

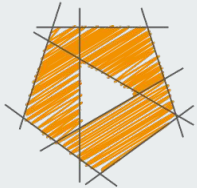


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Architecture Domains

**Everyone's favorite acronym -
BDAT**

—

**Business
Data
Application
Technology**

—

Layers of the architecture cake

Business Domain

Data Domain

Application Domain

Technology Domain

**Business
Data
Application
Technology**

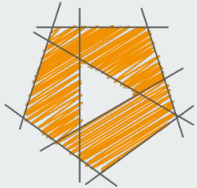
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ADM

—

The heart of TOGAF

**ADM is a process to define
enterprise architecture**

Tested and repeatable

—

Preliminary Phase

Phase A : Vision

Phases B, C, D : BDAT

—

Phases E, F : Planning

Phase G : Implementation

Phase H : Change Management

Requirements Management Phase Ongoing

**Insert Coin :
Would you like to play again?**

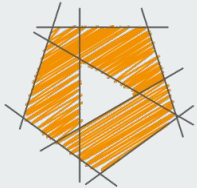
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Artifacts, Deliverables, and Building Blocks

Architecture Content Framework

Artifacts

- Lists
- Matrices
- Diagrams

Deliverables



Deliverables

“contractually specified and in turn formally reviewed, agreed, and signed off by the stakeholders”

**These are the important
documents**

Building Blocks

Architecture Building Blocks (ABBs)

**Named concepts that can be
reused in other places**

Solution Building Blocks (SBBs)

**These are implementations of
the ABBs**

ABB:

**Ability to look up a customer's
info**

—

SBB:

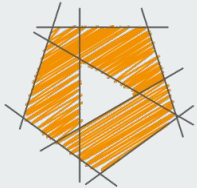
CRM Customer Search Module





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Enterprise Continuum



What's with the Star Trek references?

**A way to classify items in the
architecture repository**

**From extremely generic to
extremely specific**

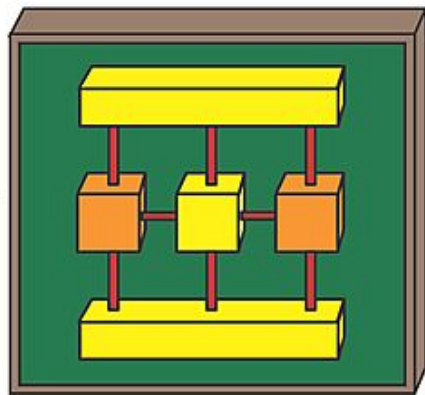
**Usually represented by a
straight line**

—

**Foundation
Architectures**



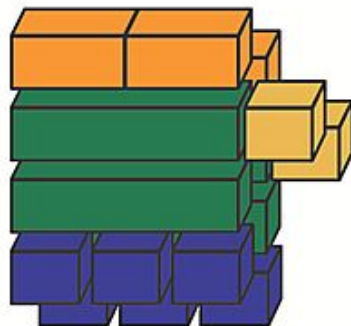
**Common
Systems
Architectures**



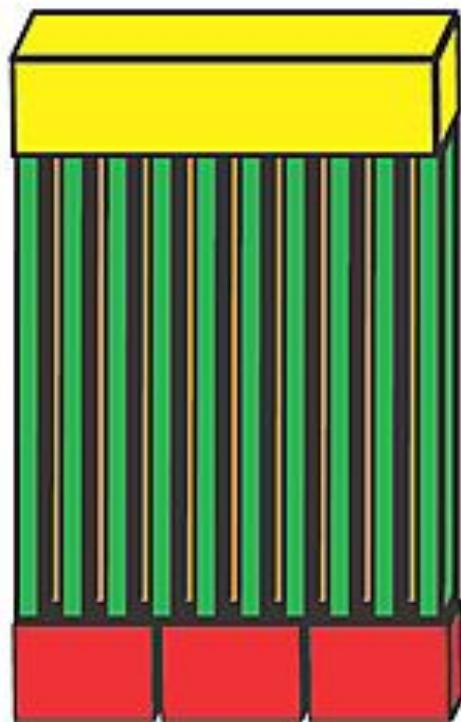
**Industry
Architectures**



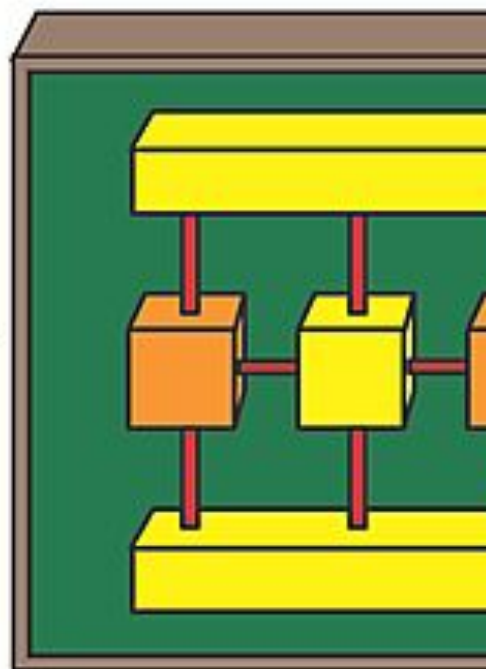
**Organization-Specific
Architectures**



Foundation Architectures



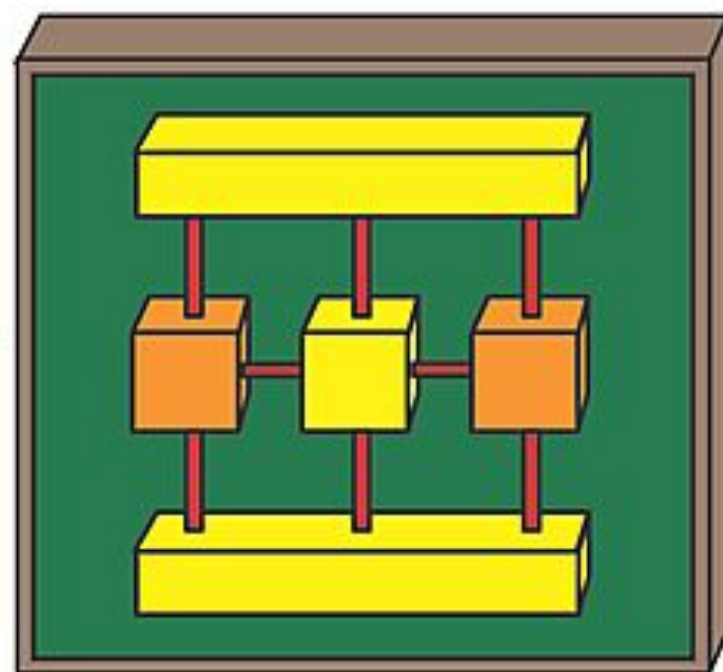
Common System Architecture



Foundation
Architectures



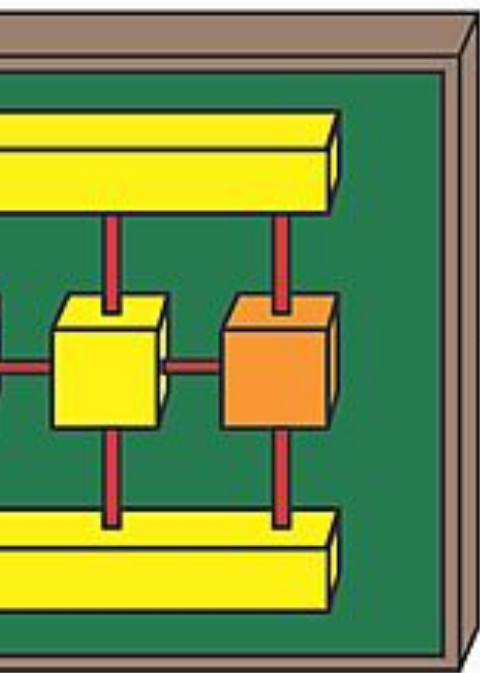
Common
Systems
Architectures



Industry
Architectures



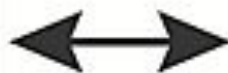
Common Systems Architectures



Industry Architectures



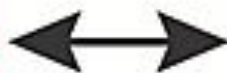
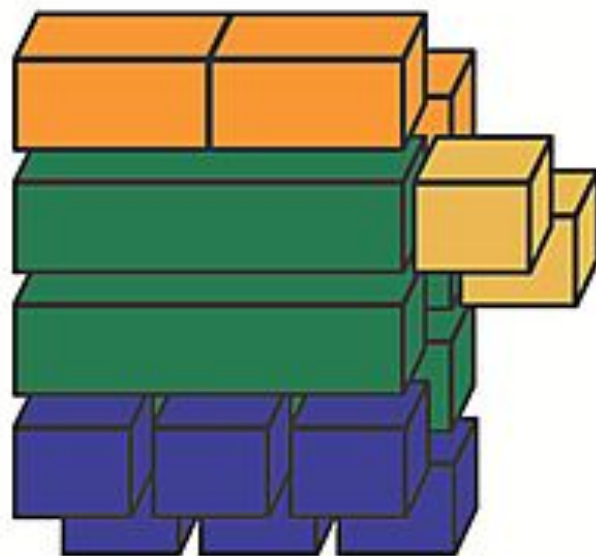
Organization-S Architectur

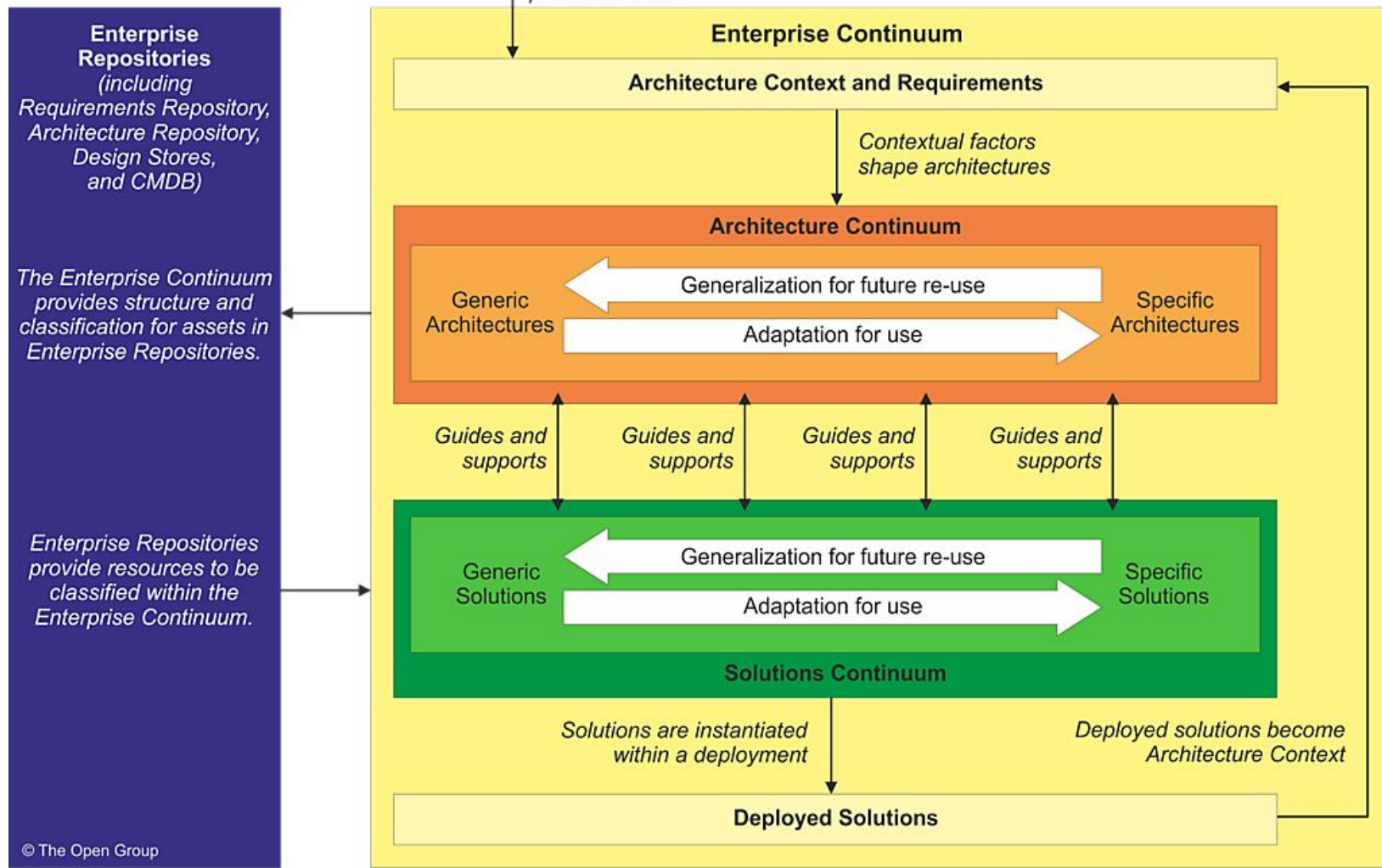


Industry Architectures



Organization-Specific Architectures



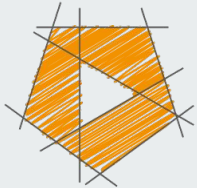






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Architecture Repository

**Central storage for important
architecture documents**

Not

C:\users\scott\Documents

A well organized file server

Versioned file system like Git

Knowledge management tool

**Each company can choose its
own tool**



The Architecture Repository contains...

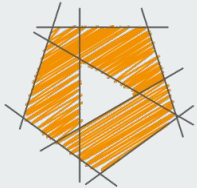
- Architecture Metamodel
- Architecture Capability
- Architecture Landscape
- Standards Library (formerly Standards Information Base (SIB))
- Reference Library
- Governance Repository (formerly Governance Log)
- Architecture Requirements Repository
- Solutions Landscape





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Architecture Capability

CAPABILITY ESTIMATION



NOT FOR THE WEAK AT HEART

An evaluation of your architecture capability

**You may have no capability
currently**

—

**Requires effort to grow
architecture capability**

Preliminary Phase of ADM

**Capability has a relationship to
governance and repository**

**Ability to establish
architecture contract part of
capability too.**

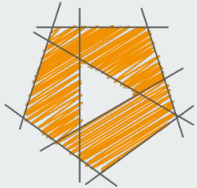
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ADM!

—

The main course

**The ADM is the process by
which an architecture is
defined**

—

10 phases

—

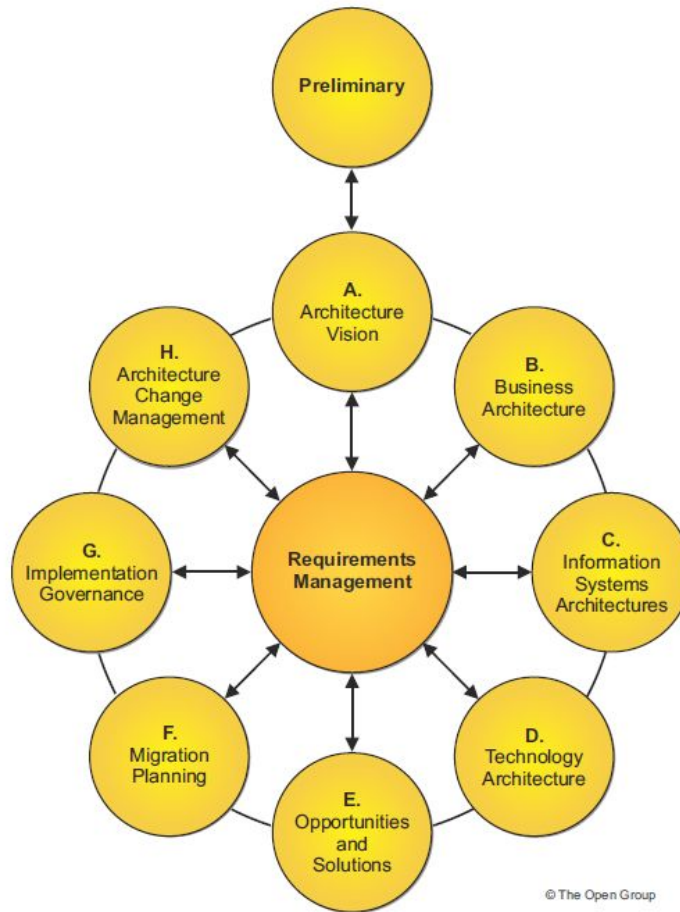


Figure 1-1 Architecture Development Cycle

8 phases make up ADM Cycle

Objective, Inputs, Steps, Outputs

**First time through might be
tough**

Repository is empty

**And low level of architecture
capability**

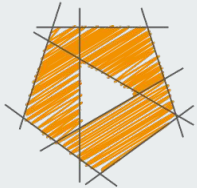
It gets easier!





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ADM Cycle

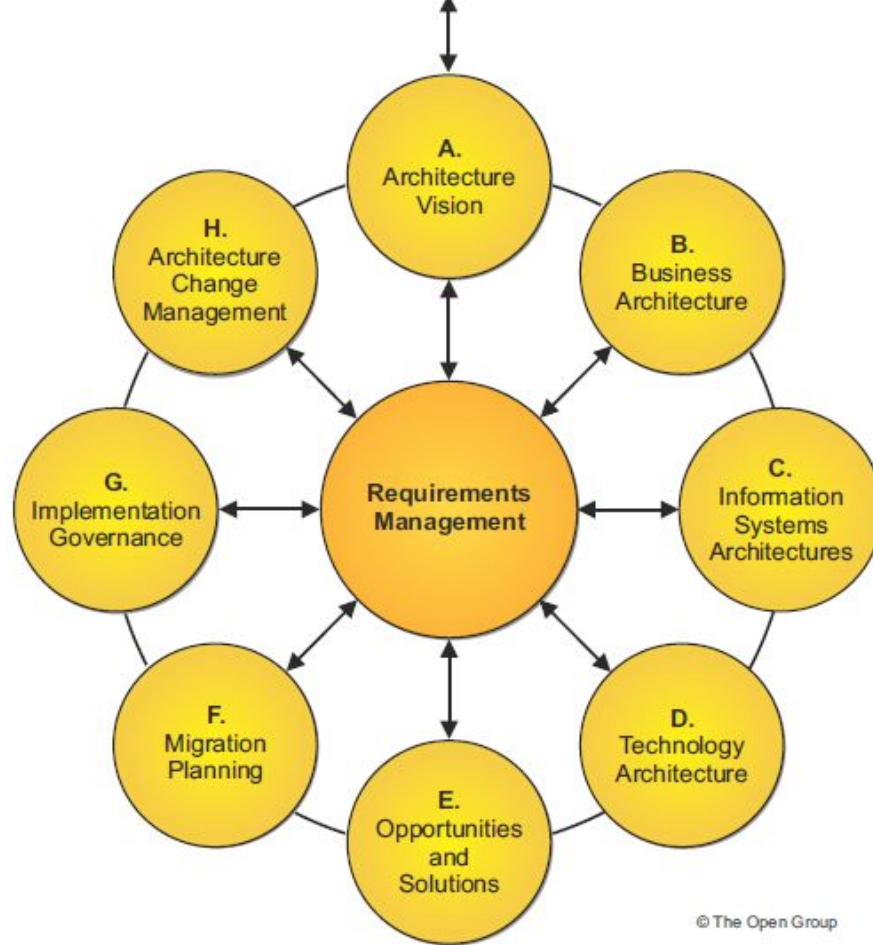


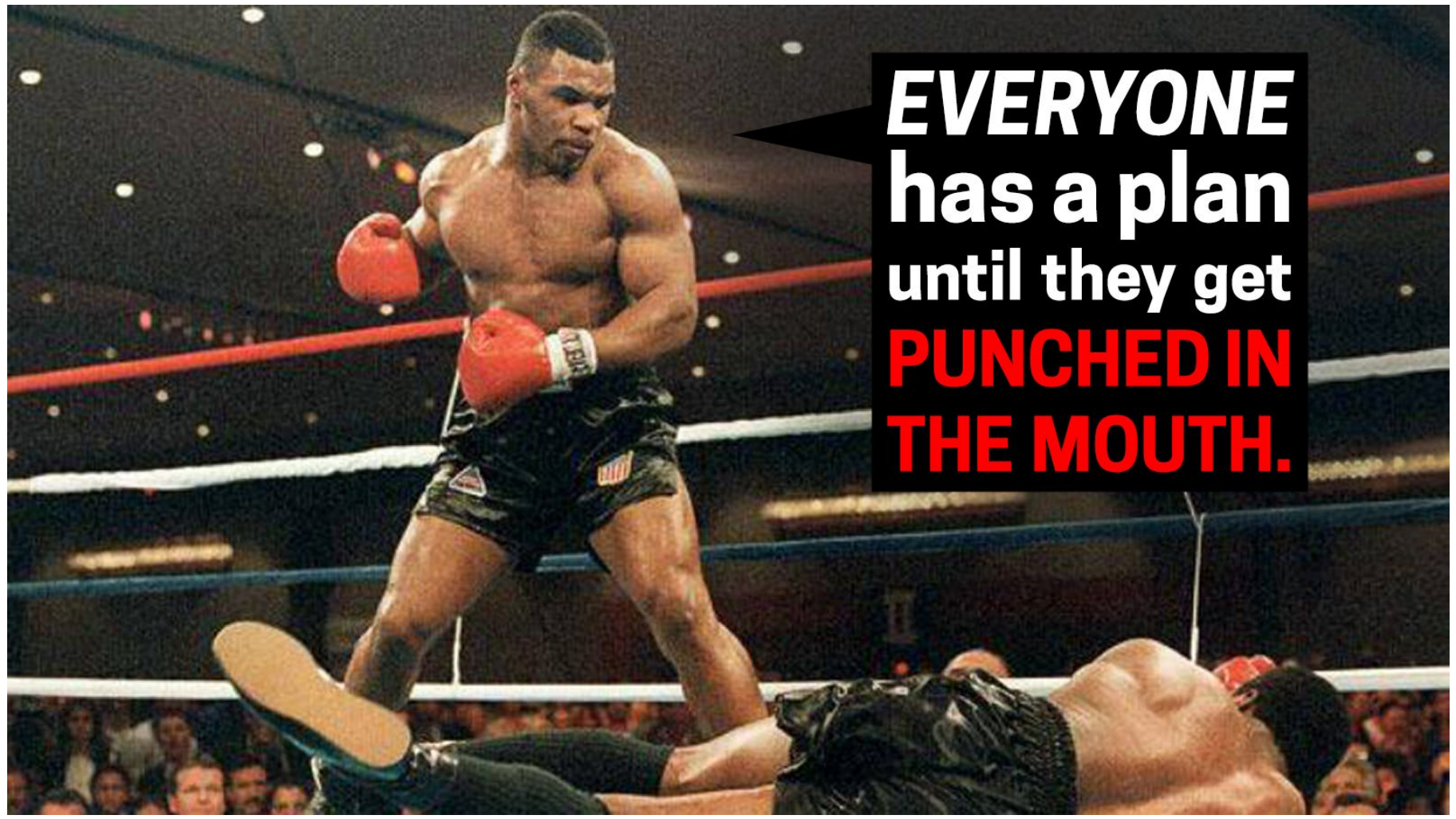
Figure 1-1 Architecture Development Cycle

8 Phases of the ADM - A thru H

ADM cycle is designed to start over once the last phase is complete

—

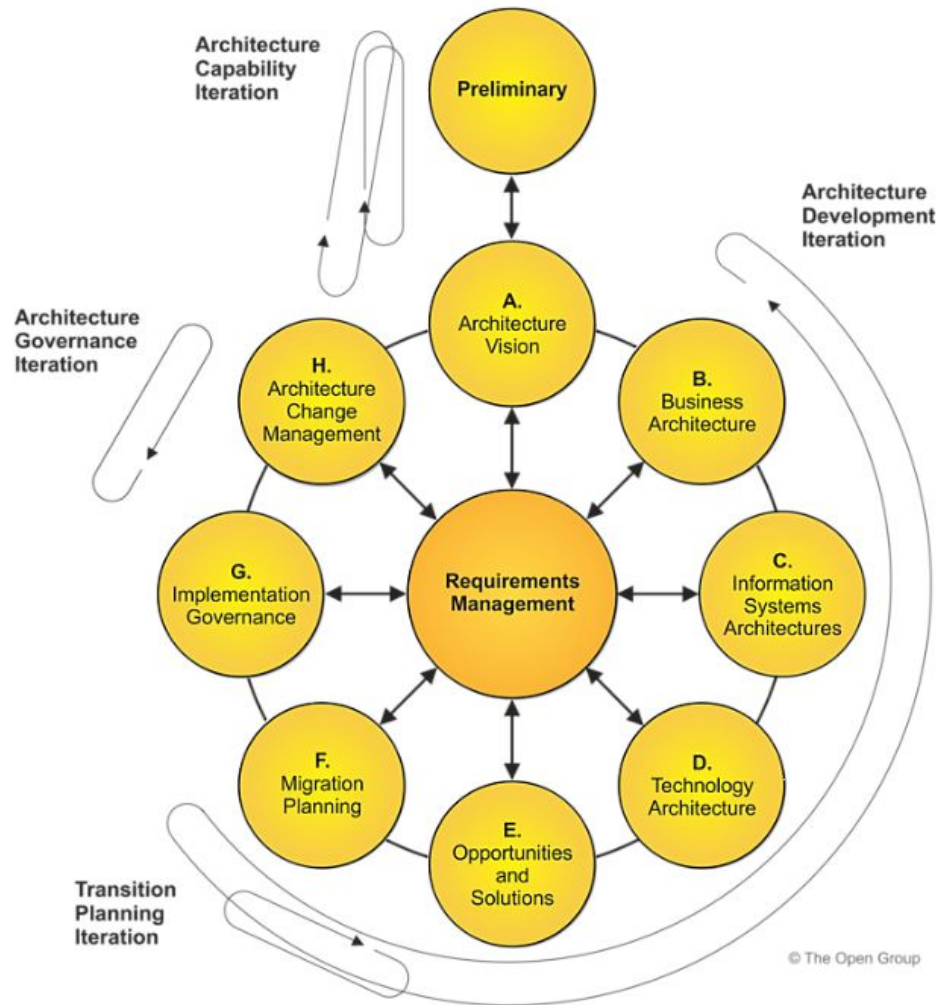
Requirements Management in the center



EVERYONE
has a plan
until they get
PUNCHED IN
THE MOUTH.

**Phases can be paused to go
back and revise a past phase**

Iterations



TOGAF designed to be tailored

**ADM can be adapted to your
needs**

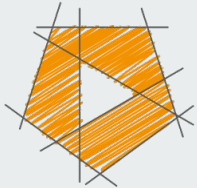


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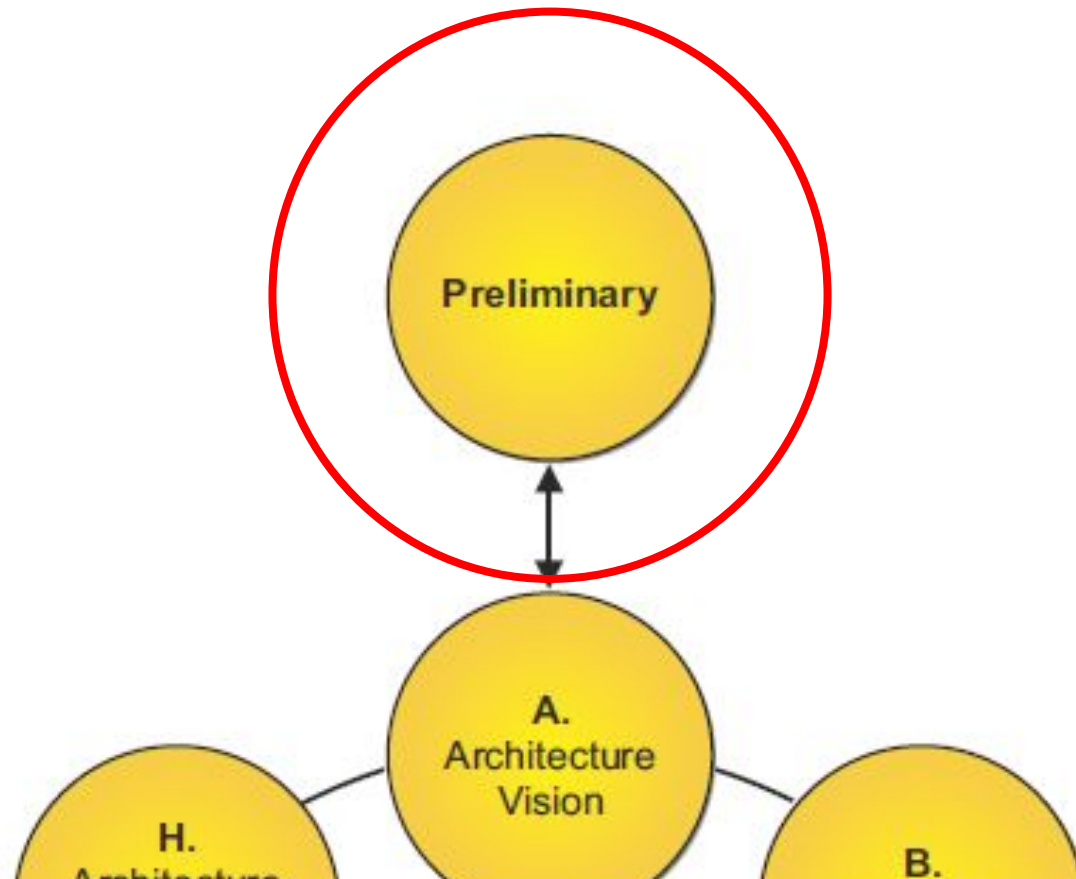
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Preliminary Phase



The purpose is to: define the
architecture capability
desired

—

The purpose is to: establish the architecture capability...

**... including architecture
principles**

**"where, what, why, who, and
how we do architecture"**



Inputs to Preliminary Phase

1. The TOGAF Library
2. Other architecture frameworks
3. Board strategies and board business plans, business strategy, IT strategy, business principles, business goals, and business drivers
4. Major frameworks operating in the business (e.g. SCRUM)
5. Governance and legal frameworks, including Architecture Governance strategy



Inputs to Preliminary Phase

6. Architecture capability
7. Partnership and contract agreements
8. Existing documents relating to architecture capability
9. Organizational Model for Enterprise Architecture
10. Existing Architecture Framework



Steps to Preliminary Phase

1. Scope the enterprise organizations impacted
2. Confirm governance and support frameworks
3. Define and establish architecture team
4. Identify and establish architecture principles
5. Tailor TOGAF and other frameworks
6. Implement architecture tools



Outputs from Preliminary Phase

1. Organizational Model for Enterprise Architecture
2. Tailored Architecture Framework
3. Initial Architecture Repository
4. Business principles, business goals, and business drivers
5. Request for Architecture Work (optional)
6. Architecture Governance Framework
7. The Architecture of the Enterprise Architecture Capability



Artifacts Produced

- Principles catalog

Principles hardly ever change

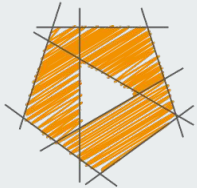
The purpose is to define and establish your architecture capability including architecture principles





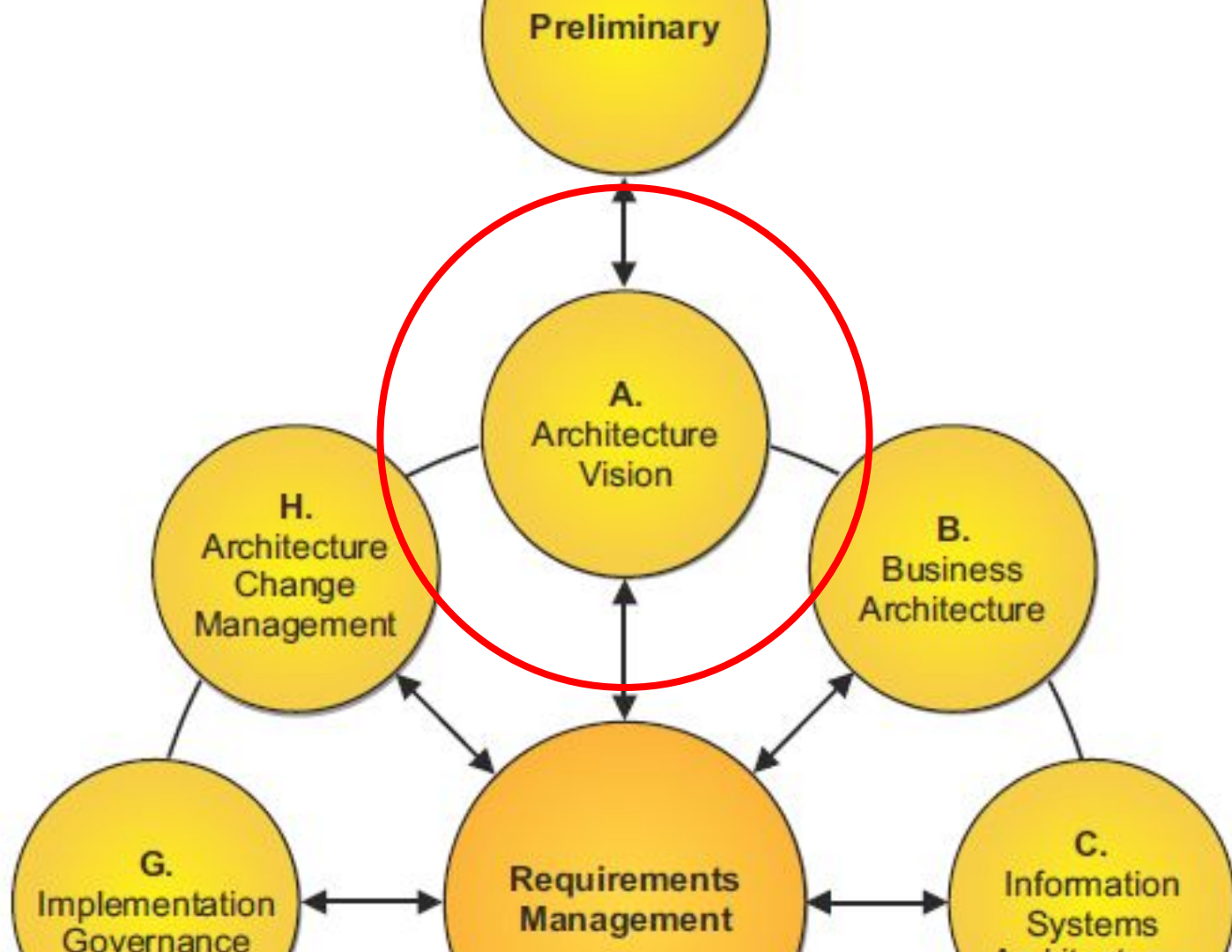
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Phase A, Architecture Vision



The purpose is to: develop a high-level aspirational vision of the business value to be delivered

**The purpose is to: obtain
approval for a Statement of
Architecture Work**

—



Inputs to Phase A

1. Architecture reference materials
2. Request for Architecture Work
3. Business principles, business goals, and business drivers
4. Organizational Model for Enterprise Architecture
5. Tailored Architecture Framework
6. Populated Architecture Repository



Steps to Phase A

1. Establish architecture project
2. Identify stakeholders, concerns and business requirements
3. Confirm business goals, drivers and constraints
4. Evaluate capabilities
5. Assess readiness for transformation
6. Define scope



Steps to Phase A

7. Confirm architecture principles, including business principles
8. Develop architecture vision
9. Define the target architecture value and KPIs
10. Identify transformation risks and mitigation activities
11. Develop Statement of Architecture Work, secure approval



Outputs from Phase A

1. Approved Statement of Architecture Work
2. Refined statements of business principles, goals and drivers
3. Architecture principles
4. Capability assessment
5. Tailored architecture framework



Outputs from Phase A

6. Architecture vision
7. Draft architecture definition document (may include baseline and target BDAT documents)
8. Communications plan
9. Additional content in the Architecture Repository



What is the Architecture Vision?

- Problem description
- Objective of the statement of architecture work
- Summary views
- Business scenario (optional)
- Refined high-level stakeholder requirements



Artifacts Produced

- Matrices: Stakeholder map matrix
- Diagrams: Business model diagram, Business capability map, Value stream map, Value chain diagram, Solution concept diagram

**The high-level vision incl
emerging technologies that
impact your industry**

—

**Very high-level, rough
description of what the
end-state of the architecture
will be**

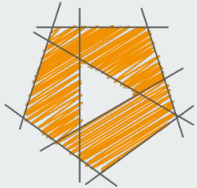
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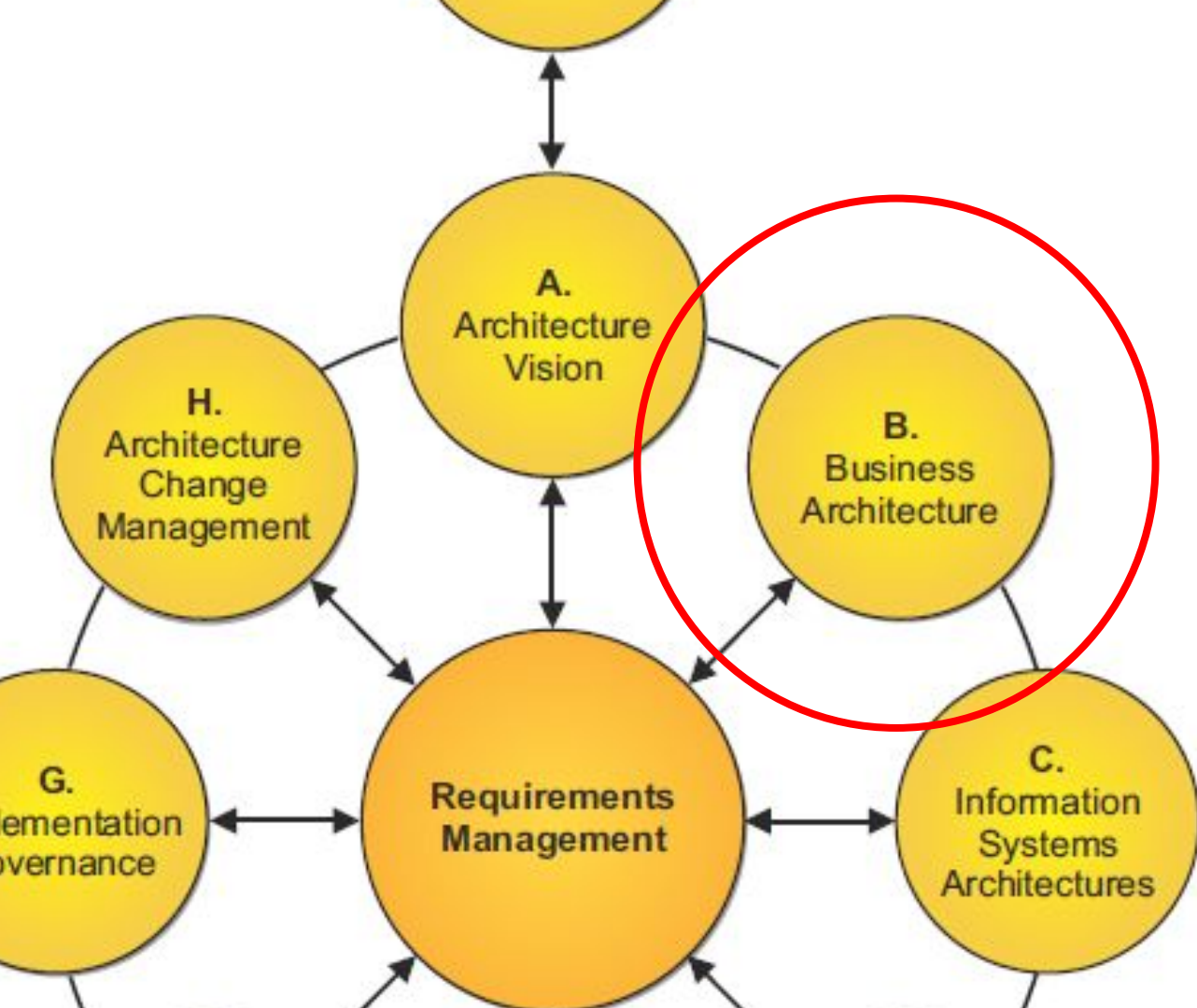
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The BDAT phases

Phase B, Business Architecture



The purpose is to:
develop the target business
architecture

The purpose is to: identify
candidate Architecture
Roadmap items based on gaps

**Based on the Architecture
Vision in the previous phase**



Inputs to Phase B

1. External reference materials
2. Request for Architecture Work (optional) (Preliminary Phase)
3. Business principles, goals, and drivers (Phase A)
4. Capability assessment (Phase A)
5. Communications plan (Phase A)



Inputs to Phase B

6. Organization Model for Enterprise Architecture (Prelim Phase)
7. Tailored Architecture Framework (Prelim Phase)
8. Approved Statement of Architecture Work (Phase A)
9. Architecture principles (Phase A)
10. Enterprise continuum
11. Architecture Repository



Inputs to Phase B

12. Architecture vision (Phase A)
13. Draft architecture definition document
(draft version of all baseline and target BDAT documents)
(Phase A)



Steps for Phase B

1. Select reference models, viewpoints, and tools
2. Develop baseline business architecture description
3. Develop target business architecture description
4. Perform gap analysis
5. Define candidate roadmap components
6. Resolve impacts across the architecture landscape



Steps for Phase B

7. Conduct formal stakeholder review
8. Finalize business architecture
9. Create architecture definition document



Outputs from Phase B

1. Refined Phase A deliverables
2. Draft architecture definition document
 - a. Baseline business architecture, approved version (detailed)
 - b. Target business architecture, approved version (detailed)
3. Draft architecture requirements specification – gap analysis
4. Business architecture components of an architecture roadmap



Artifacts Produced

Catalogs: Value Stream catalog, Business Capabilities catalog, Value Stream Stages catalog, Organization/actor catalog, Driver/goal/objective catalog, role catalog, business service/function catalog, location catalog, process/event/control/product catalog, contract/measure catalog

Matrices: Value Stream/Capability matrix, Strategy/Capability matrix, Capability/Organization matrix, Business interaction matrix, actor/role matrix

Diagrams: Business Model diagram, Business Capability Map, Value Stream Map, Organization Map, Business footprint diagram, business service/information diagram, function decomposition diagram, product lifecycle diagram, goal/objective/service diagram, business use-case diagram, organization decomposition diagram, process flow diagram, event diagram

**A: Architecture Vision about
establishing the business goals**

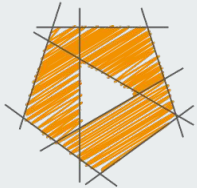
**B: Business architecture about
how to get there**





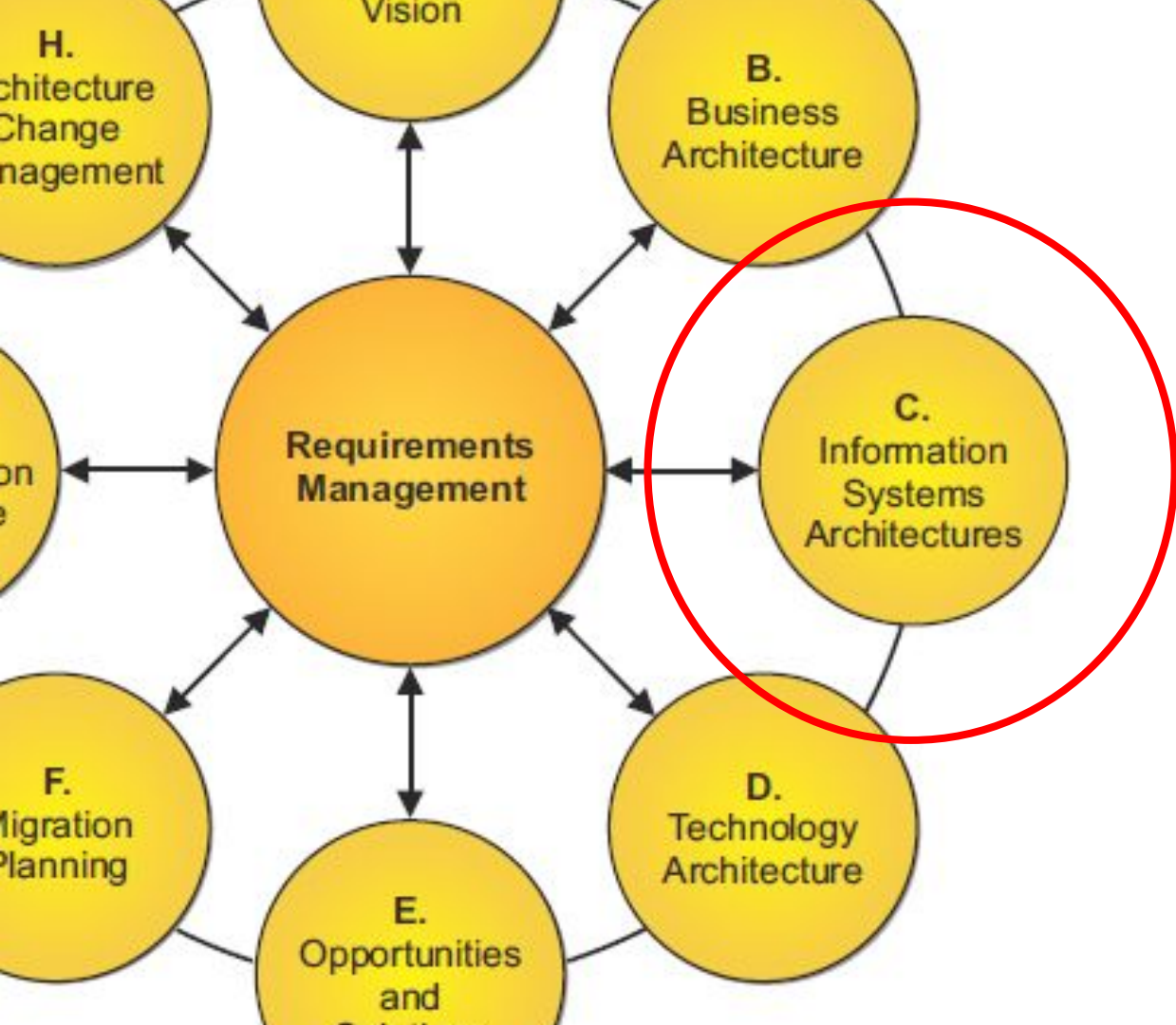
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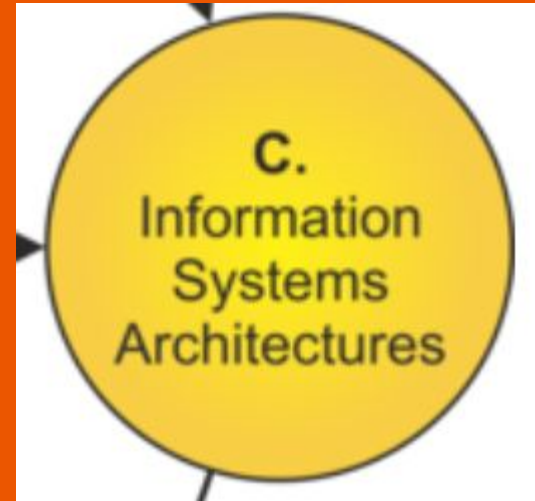
Phase C



Phase C is an odd duck



**2 phases in 1:
Define data and
application
architectures**



**The objective, inputs, steps,
output are almost the same**

**The purpose is to: develop the
target information systems
architectures**

—

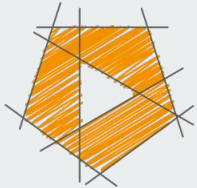
The purpose is to: identify
candidate Architecture
Roadmap items based on gaps





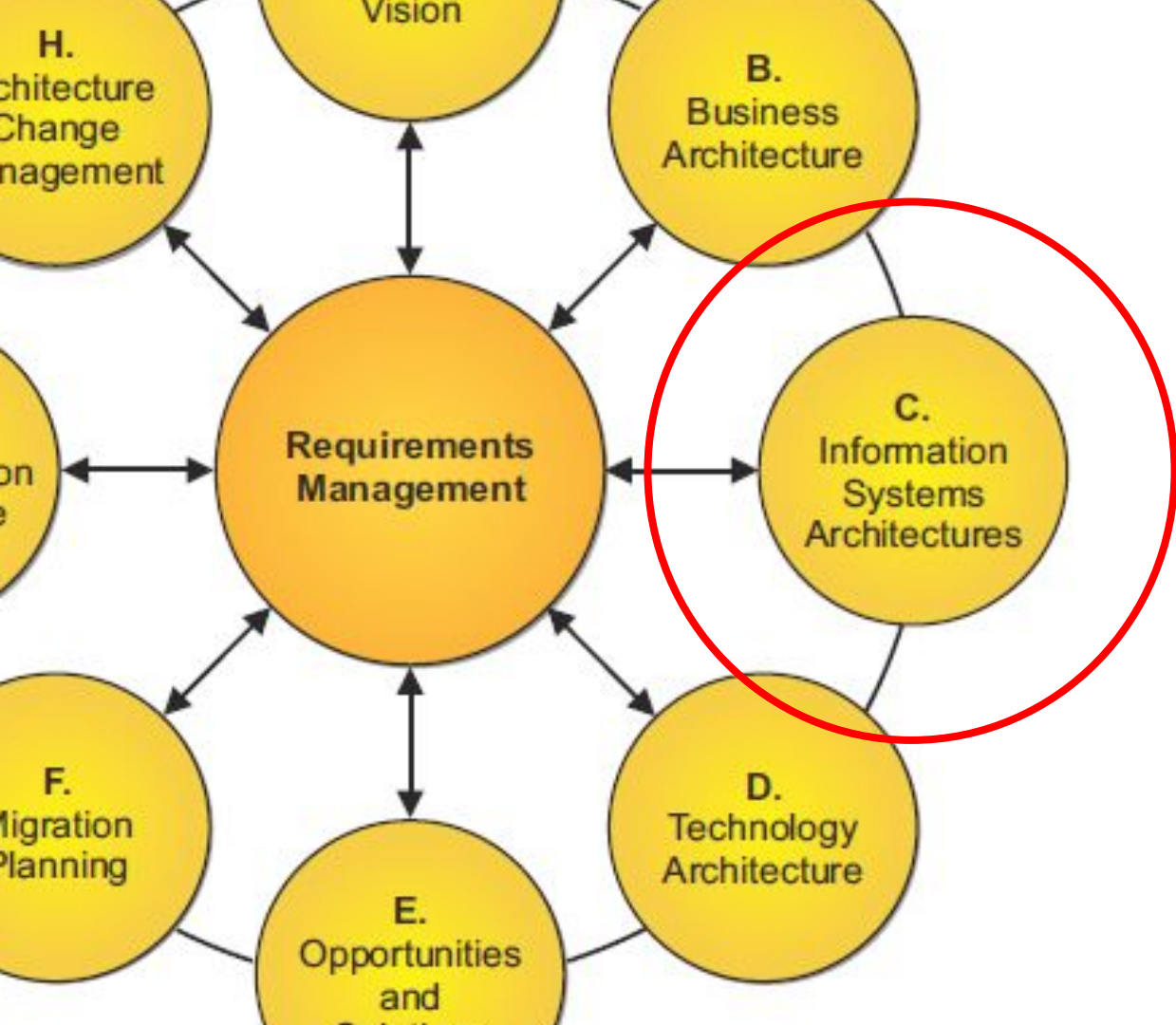
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Phase C, Data Architecture



**The purpose is to: develop the
target data architecture**

—

The purpose is to: identify
candidate Architecture
Roadmap items based on gaps



Inputs to Phase C, Data Architecture

1. External reference materials
2. TOGAF® Series Guide: Information Architecture - Customer Master Data Management
3. Request for Architecture Work (optional) (Preliminary Phase)
4. Capability assessment (Phase A)
5. Communications plan (Phase A)
6. Organization Model for Enterprise Architecture (Prelim Phase)
7. Tailored Architecture Framework (Prelim Phase)



Inputs to Phase C, Data Architecture

7. Data principles (The TOGAF Standard: ADM Techniques)
8. Approved Statement of Architecture Work (Phase A)
9. Architecture vision (Phase A)
10. Architecture Repository
11. Draft architecture documents and draft requirements (Phase B)
12. Business related architecture roadmap (Phase B)



Steps for Phase C, Data Architecture

1. Select reference models, viewpoints, and tools
2. Develop baseline data architecture description
3. Develop target data architecture description
4. Perform gap analysis
5. Define candidate roadmap components
6. Resolve impacts across the architecture landscape



Steps for Phase C, Data Architecture

7. Conduct formal stakeholder review
8. Finalize data architecture
9. Create architecture definition document



Outputs from Phase C, Data Architecture

1. Refined Phase A deliverables
2. Draft architecture definition document
 - a. Baseline data architecture, approved version
 - b. Target data architecture, approved version
3. Draft architecture requirements specification – gap analysis
4. Data architecture components of an architecture roadmap



Artifacts Produced

Catalogs: Data entity/data component catalog

Matrices: Data entity/business function matrix, application/data matrix

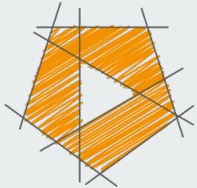
Diagrams: Conceptual data diagram, logical data diagram, data dissemination diagram, data security diagram, data migration diagram, data lifecycle diagram





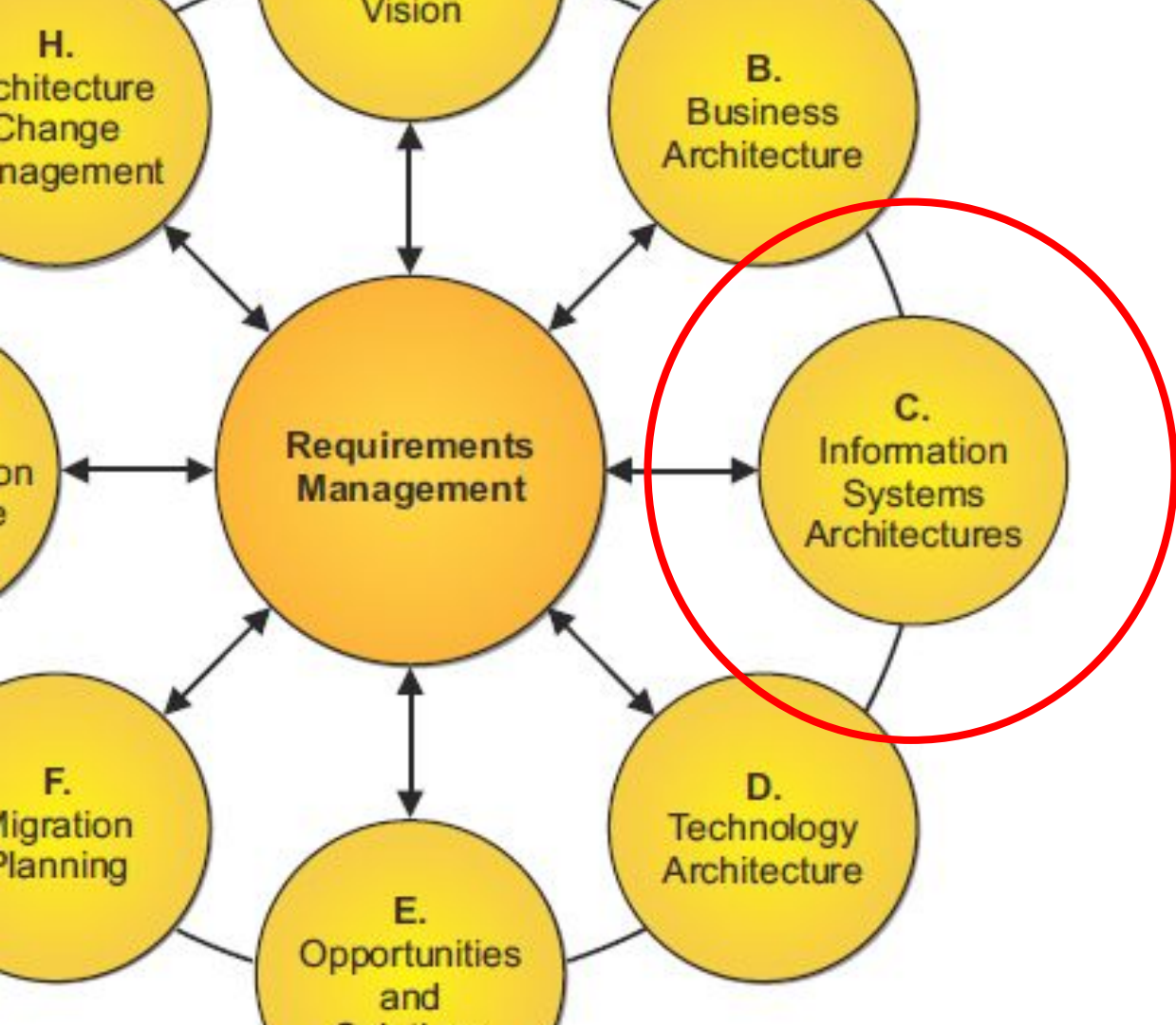
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Phase C, Application Architecture



**The purpose is to: develop the
target application architecture**

—

**The purpose is to: identify
candidate Architecture
Roadmap items based on gaps**



Inputs to Phase C, Application Architecture

1. External reference materials
2. Request for Architecture Work (optional) (Preliminary Phase)
3. Capability assessment (Phase A)
4. Communications plan (Phase A)
5. Organization Model for Enterprise Architecture (Prelim Phase)
6. Tailored Architecture Framework (Prelim Phase)



Inputs to Phase C, Application Architecture

7. Application principles (The TOGAF Standard: ADM Techniques)
8. Approved Statement of Architecture Work (Phase A)
9. Architecture vision (Phase A)
10. Architecture Repository
11. Draft architecture documents and draft requirements (Phase B)
12. Business and data related architecture roadmap (Phase B)



Steps for Phase C, Application Architecture

1. Select reference models, viewpoints, and tools
2. Develop baseline application architecture description
3. Develop target application architecture description
4. Perform gap analysis
5. Define candidate roadmap components
6. Resolve impacts across the architecture landscape



Steps for Phase C, Application Architecture

7. Conduct formal stakeholder review
8. Finalize application architecture
9. Create architecture definition document



Outputs from Phase C, Application Architecture

1. Refined Phase A deliverables
2. Draft architecture definition document
 - a. Baseline application architecture, approved version
 - b. Target application architecture, approved version
3. Draft architecture requirements specification – gap analysis
4. Application architecture components of an architecture roadmap



Artifacts Produced

Catalogs: Application portfolio catalog, interface catalog

Matrices: Application/organization matrix, role/application matrix, application function matrix, application interaction matrix

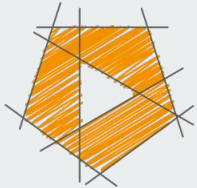
Diagrams: Application Communication diagram, Application and User Location diagram, Application Use-Case diagram, Enterprise Manageability diagram, Process/Application Realization diagram, Software Engineering diagram, Application Migration diagram, Software Distribution diagram





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Phase D, Technology Architecture

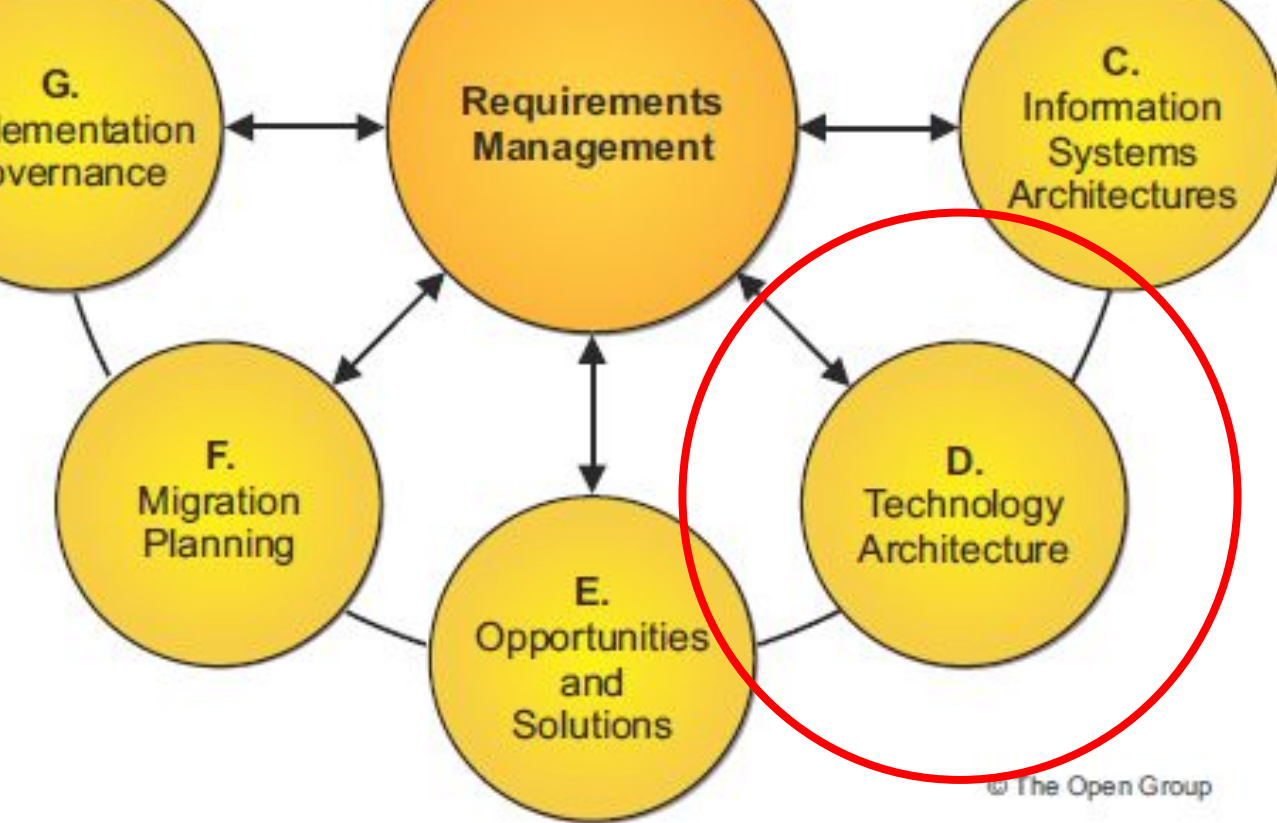


Figure 1-1 Architecture Development Cycle

**The purpose is to: develop the
target technology architecture**

—

The purpose is to: identify
candidate Architecture
Roadmap items based on gaps



Inputs to Phase D

1. External reference materials
2. Product information on candidate products
3. Request for Architecture Work (optional) (Prelim Phase)
4. Capability assessment (Phase A)
5. Communications plan (Phase A)
6. Organization Model for Enterprise Architecture (Prelim Phase)
7. Tailored Architecture Framework (Prelim Phase)



Inputs to Phase D

8. Technology principles (The TOGAF Standard: ADM Techniques)
9. Approved Statement of Architecture Work (Phase A)
10. Architecture vision (Phase A)
11. Architecture Repository
12. Draft architecture documents and draft requirements (Phase B and C)
13. Business, data and application components of an architecture roadmap (Phase B and C)



Steps for Phase D

1. Select reference models, viewpoints, and tools
2. Develop baseline technology architecture description
3. Develop target technology architecture description
4. Perform gap analysis
5. Define candidate roadmap components
6. Resolve impacts across the architecture landscape



Steps for Phase D

7. Conduct formal stakeholder review
8. Finalize technology architecture
9. Create architecture definition document



Outputs from Phase D

1. Refined Phase A deliverables
2. Draft architecture definition document
 - a. Baseline technology architecture, approved version
 - b. Target technology architecture, approved version
3. Draft architecture requirements specification – gap analysis
4. Technology architecture components of an architecture roadmap



Artifacts Produced

Catalogs: Technology Standards catalog, Technology Portfolio catalog

Matrices: Application/Technology matrix

Diagrams: Environments and Locations diagram, Platform Decomposition diagram, Processing diagram, Networked Computing/Hardware diagram, Communications Engineering diagram

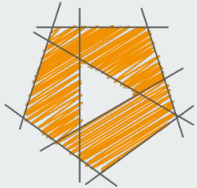
**Look for opportunities using
emerging technologies**





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Phase E, Opportunities and Solutions



Figure 1-1 Architecture Development Cycle

Congratulations!

A meme featuring Leonardo DiCaprio from the movie Inception, dressed in a tuxedo and holding a glass of champagne, smiling and celebrating. The background is a blurred party scene with confetti and lights.

WE DID IT

TIME TO CELEBRATE

The purpose is to:
generate the initial
Architecture Roadmap

The purpose is to:
identify if transition
architectures are required

The purpose is to:
define solution building blocks
(SBBs)



Inputs to Phase E

1. External reference materials
2. Product information
3. Request for Architecture Work (optional) (Prelim Phase)
4. Capability assessment (Phase A)
5. Communications plan (Phase A)
6. Planning methodologies
7. Organization Model for Enterprise Architecture (Prelim Phase)
8. Governance models and frameworks



Inputs to Phase E

9. Tailored Architecture Framework (Prelim Phase)
10. Approved Statement of Architecture Work (Phase A)
11. Architecture vision (Phase A)
12. Architecture Repository
13. Draft architecture documents and draft requirements (BDAT)
14. Change requests for existing business programs and projects
15. Candidate architecture roadmap components



Steps for Phase E

1. Determine key corporate change attributes
2. Determine business constraints for implementation
3. Review and consolidate gap analysis from Phases B to D
4. Review consolidated requirements across business functions
5. Consolidate and reconcile interoperability requirements
6. Refine and validate dependencies



Steps for Phase E

7. Confirm readiness and risk for business transformation
8. Formulate implementation and migration strategy
9. Identify and group major work packages
10. Identify transition architectures
11. Create the architecture roadmap & implementation and migration plan



Outputs from Phase E

1. Refined Phase A deliverables
2. Draft architecture definition document
 - a. incl. baseline and target BDAT architectures, approved
 - b. incl. transition architectures, views
3. Draft architecture requirements specification – incl. gap analysis
4. Capability assessments
5. Architecture roadmap
 - a. incl. work package portfolio
6. Implementation and migration plan, draft version



Artifacts Produced

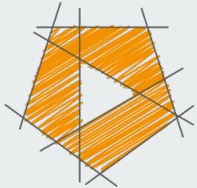
Diagrams: Product context diagram, benefits diagram





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Phase F, Migration Planning

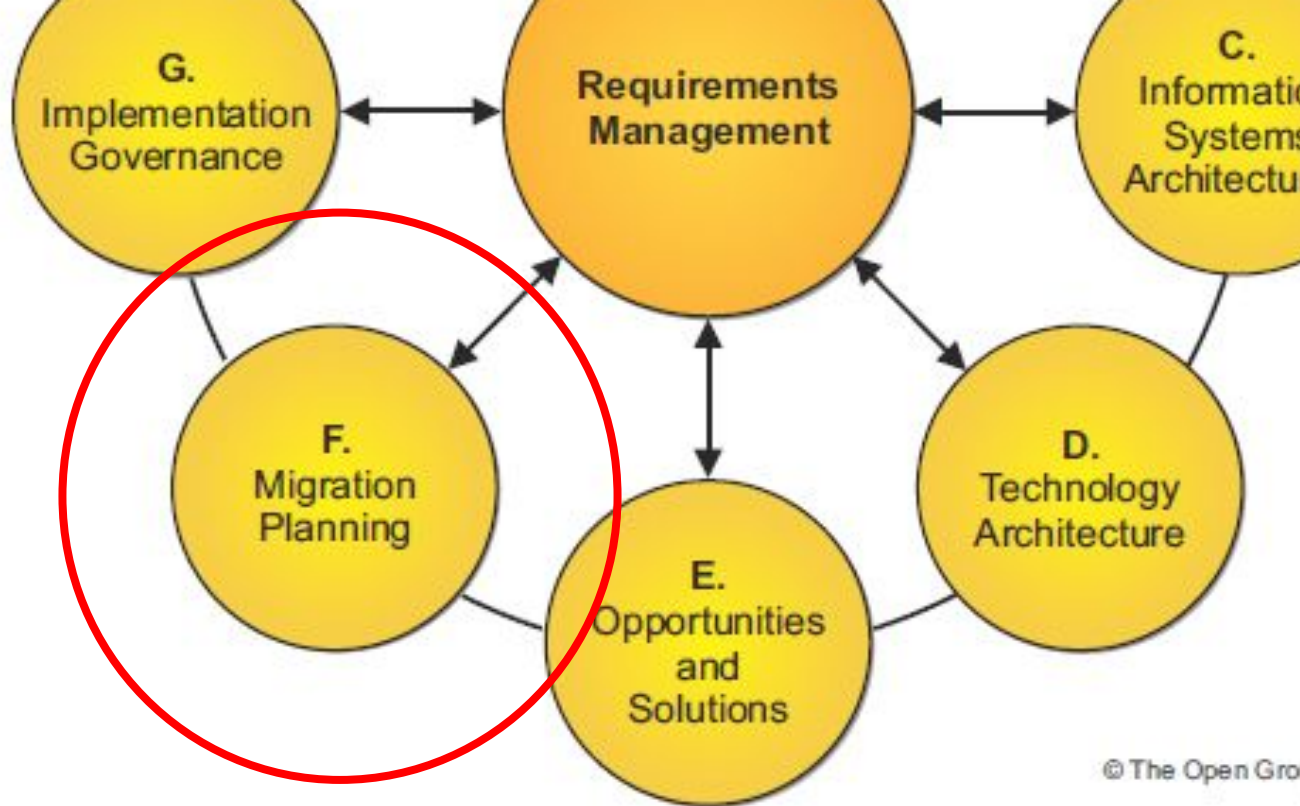


Figure 1-1 Architecture Development Cycle

The purpose is to:
finalize the architecture
roadmap and migration plan

The purpose is to:
ensure migration plan is
aligned with enterprise
approach to change

—

The purpose is to:
ensure business value and cost
of work packages is
understood



Inputs to Phase F

1. External reference materials
2. Request for Architecture Work (optional) (Prelim Phase)
3. Capability assessment (Phase A)
4. Communications plan (Phase A)
5. Organization Model for Enterprise Architecture (Prelim Phase)
6. Governance Models and Frameworks
7. Tailored Architecture Framework (Prelim Phase)



Inputs to Phase F

8. Approved Statement of Architecture Work (Phase A)
9. Architecture vision (Phase A)
10. Architecture Repository
11. Draft architecture documents and draft requirements (BDAT)
12. Change requests for existing business programs and projects
13. Architecture roadmap, draft version (Phase E)
14. Capability Assessment
15. Implementation and migration plan , draft version (Phase E)



Steps for Phase F

1. Confirm management framework interactions
2. Assign a business value to each work package
3. Estimate resource requirements, project timings, etc
4. Prioritize the migration projects via cost/benefit assessment and risk validation
5. Confirm architecture roadmap and update architecture definition document
6. Generate implementation and migration plan
7. Complete the architecture development cycle, lessons learned



Outputs from Phase F

1. Implementation and Migration Plan, approved version
2. Finalized architecture definition document
3. Finalized architecture requirements
4. Finalized architecture roadmap
5. Reusable ABBs
6. Requests for Architecture Work for next ADM cycle (if any)
7. Implementation of governance model
8. Change requests for architecture capability from lessons learned



Artifacts Produced

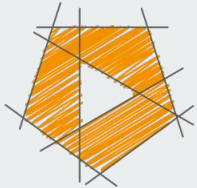
None





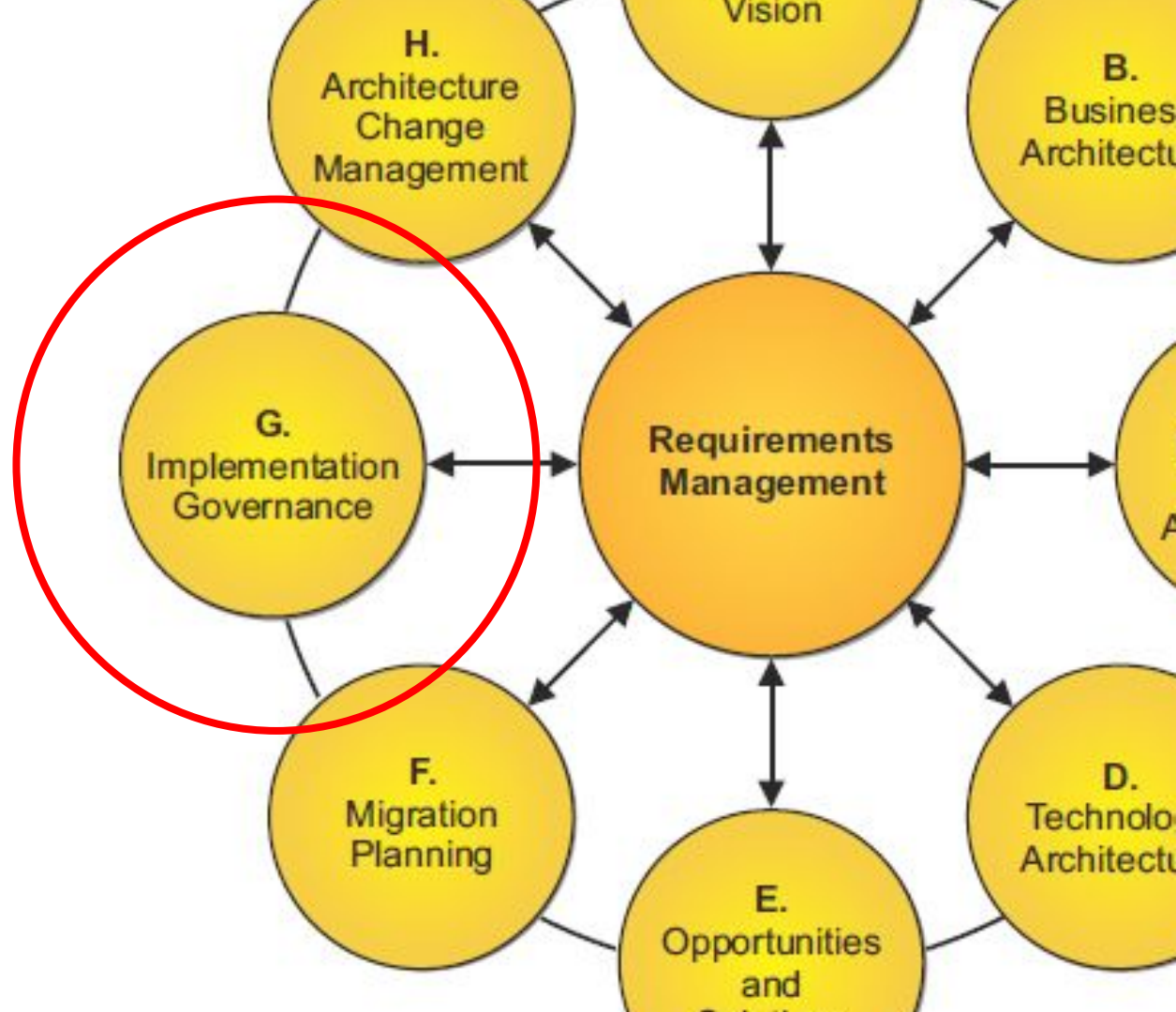
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Phase G, Implementation Governance



The purpose is to:
ensure conformance

The purpose is to:
perform architecture
governance



Inputs to Phase G

1. External reference materials
2. Request for Architecture Work (optional) (Prelim Phase)
3. Capability assessment (Phase A)
4. Organization Model for Enterprise Architecture (Prelim Phase)
5. Tailored Architecture Framework (Prelim Phase)
6. Approved Statement of Architecture Work (Phase A)
7. Architecture vision (Phase A)



Inputs to Phase G

8. Architecture Repository
9. Architecture definition documents and requirements (BDAT)
10. Architecture roadmap, approved version (Phase F)
11. Implementation governance model
12. Architecture contract
13. Request for architecture work for next ADM cycle (Phase F)
14. Implementation and migration plan , approved version (Phase F)



Steps for Phase G

1. Confirm Scope and Priorities for Deployment
2. Identify Deployment Resources and Skills
3. Guide Development of Solutions Deployment
4. Perform Enterprise Architecture Compliance Reviews
5. Implement Business and IT Operations
6. Perform Post-Implementation Review and Close the Implementation



Outputs from Phase G

1. Architecture contract

“...joint agreements between development partners and sponsors on the deliverables, quality, and fitness-for-purpose of an architecture”

Read TOGAF Fundamental Content specification

2. Compliance assessments
3. Change requests
4. Architecture-compliant solutions deployed



Artifacts Produced

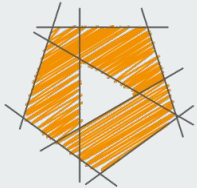
None





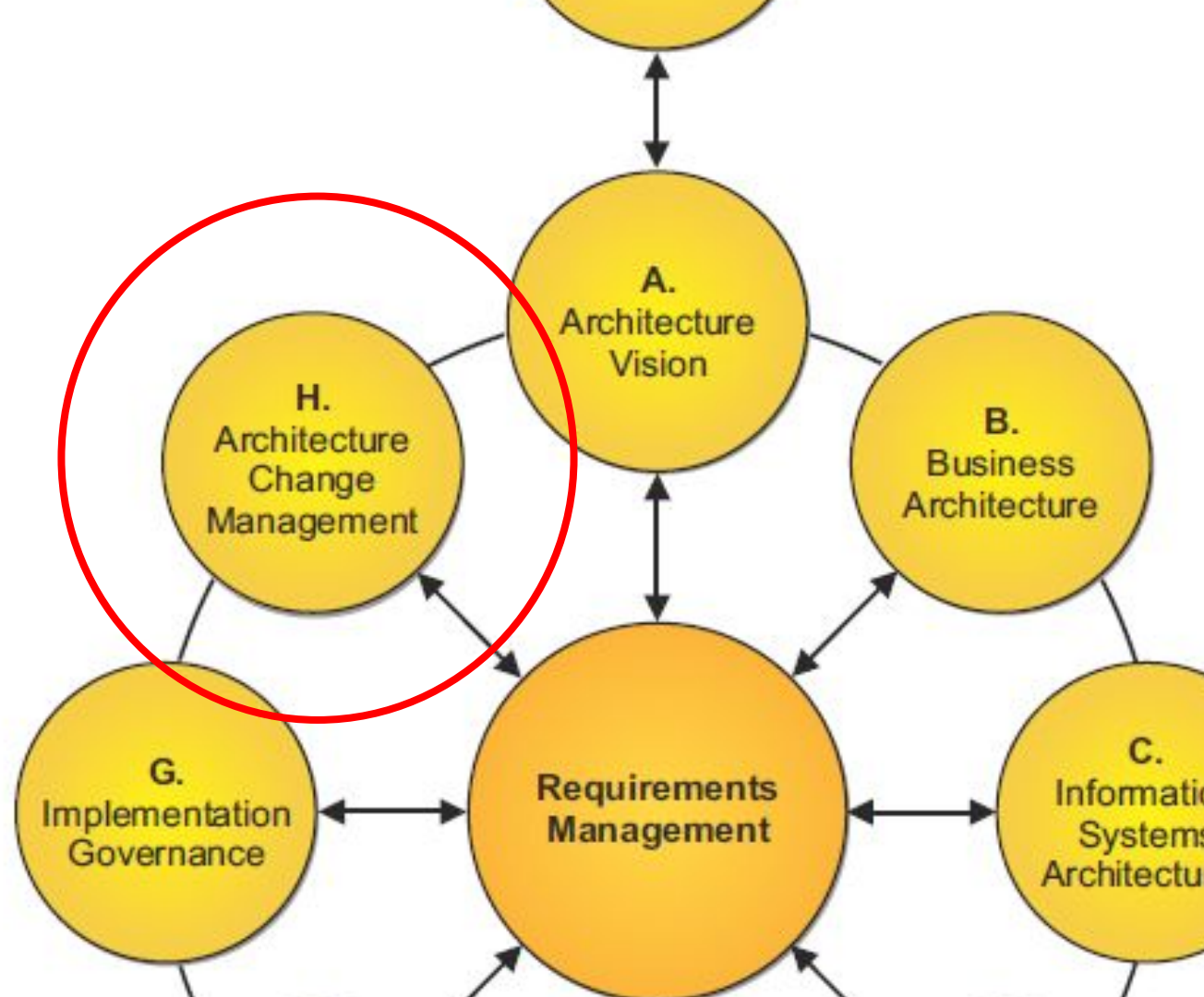
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Phase H, Architecture Change Management



The purpose is to:
maintain architecture lifecycle

The purpose is to:
execute architecture
governance

The purpose is to:
maintain architecture
capability



Inputs to Phase H

1. External reference materials
2. Request for Architecture Work (optional) (Prelim Phase)
3. Organization Model for Enterprise Architecture (Prelim Phase)
4. Tailored Architecture Framework (Prelim Phase)
5. Statement of Architecture Work (Phase A)
6. Architecture vision (Phase A)
7. Architecture Repository



Inputs to Phase H

8. Architecture definition documents and requirements (BDAT)
9. Architecture roadmap (Phase F)
10. Change requests – technology changes, business changes, lessons learned
11. Implementation governance model
12. Architecture contract (Phase G)
13. Compliance assessments (Phase G)
14. Implementation and migration plan , approved version (Phase F)



Steps for Phase H

1. Establish Value Realization Process
2. Deploy Monitoring Tools
3. Manage Risks
4. Provide Analysis for Architecture Change Management
5. Develop Change Requirements to Meet Performance Targets
6. Manage Governance Process
7. Activate the Process to Implement Change



Outputs from Phase H

1. Architecture updates (maintenance)
2. Change to architecture framework and principles (maintenance)
3. New Request for Architecture Work (for major changes)
4. Statement of architecture work (updated if necessary)
5. Architecture contract (updated if necessary)
6. Compliance assessments (updated if necessary)



Artifacts Produced

None

**Does a proposed change
require a new ADM cycle?**

—

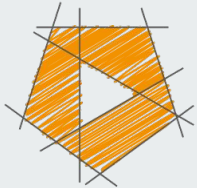
**Or support the change as part
of change management?**





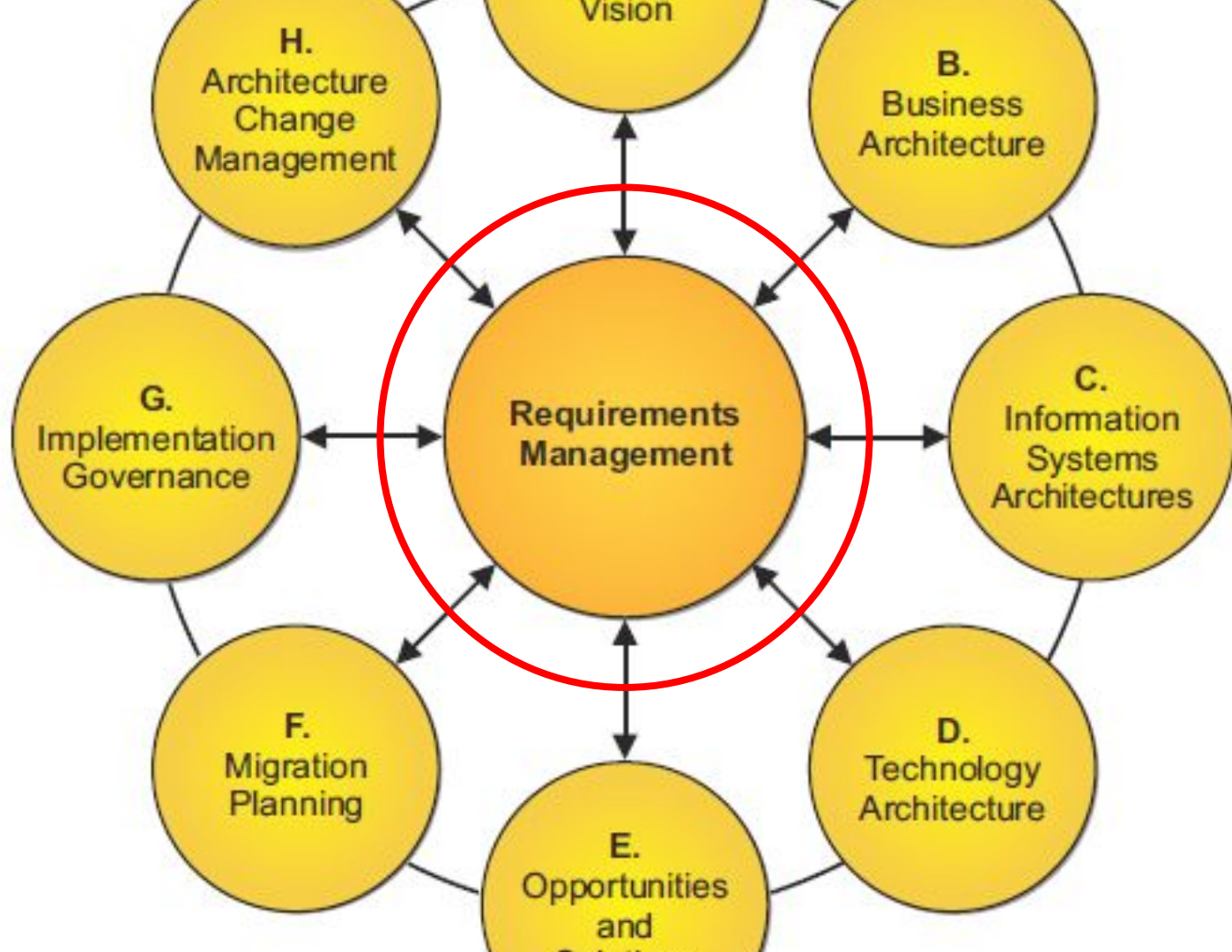
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Requirements Management



Technically, it's a phase. But it's constant.

**Continuous process of
handling change during any
phase of the ADM process.**

—

Being able to handle change at any time is crucial for the ADM to succeed.

The purpose is:
ensure the process is
sustained through all phases

**The purpose is:
manage change while the ADM
cycle is in progress**

**The purpose is:
provide the requirements to
each ADM phase**

**The RM Phase manages the
“flow” of requirements**

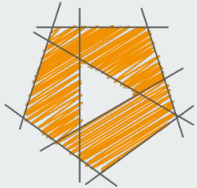
**Requirements Management is
the process that manages the
repository**





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ADM Techniques



ADM Guidelines and Techniques

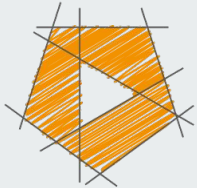
- Architecture Principles
- Stakeholder Management
- Architecture Patterns
- Gap Analysis
- Migration Planning Techniques
- Interoperability Requirements
- Business Transformation Readiness Assessment
- Risk Management
- Architecture Alternatives and Trade-Offs





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Architecture Principles

**An enduring set of general
rules and guidelines about
architecture**

**Not intended to change very
often**

**Defined during the Preliminary
Phase**

**Serves as a statement by
which you can use to make
decisions**

—

Example - Data Architecture Principle

“Data Trustee”

“Each data element has a trustee accountable for data quality.”

—



Five Elements of a Good Principle

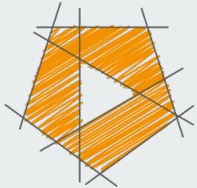
- Understandability
- Robustness
- Completeness
- Consistency
- Stability





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Stakeholder Management

**“An individual, team,
organization, or class thereof,
having an interest in a system.”**

—

**As an architect, being able to
win support for your plans is a
skill.**

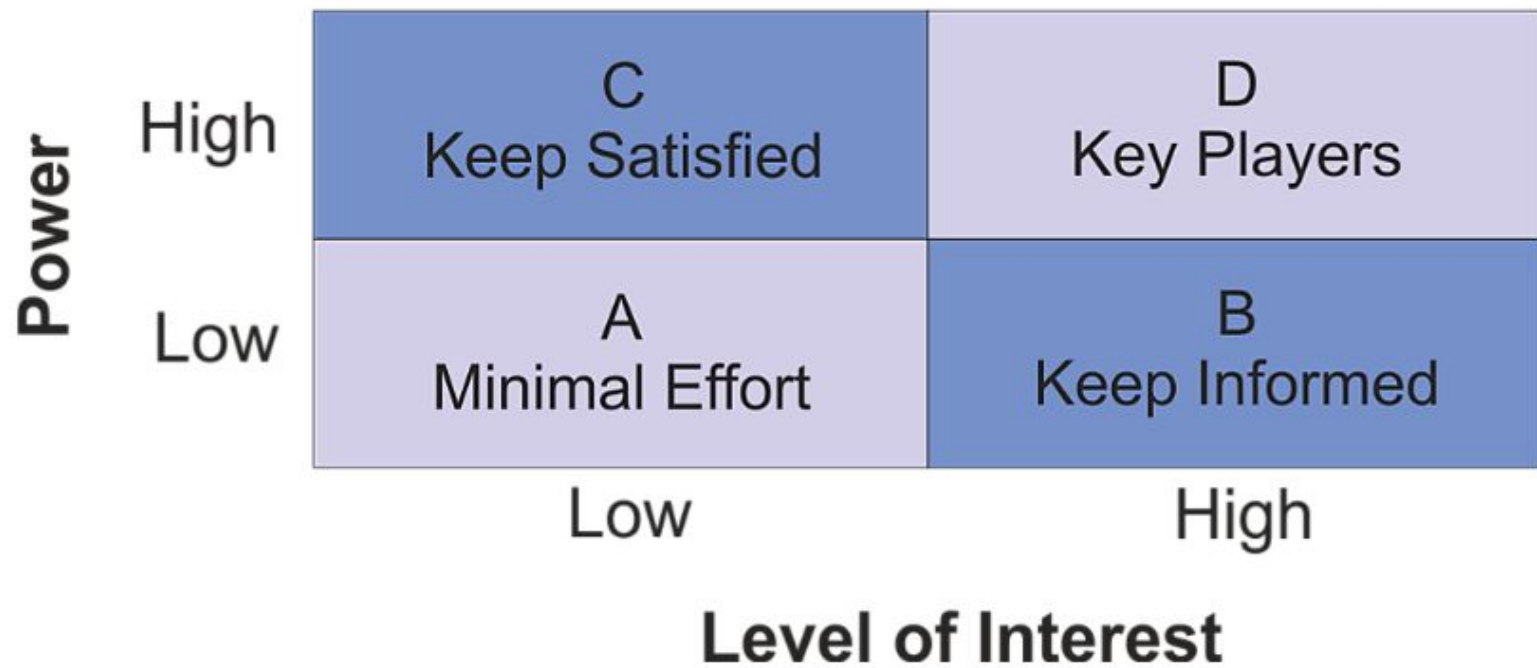
**Knowing who the most
powerful stakeholders are...**

**Makes things easier for
approval and budgets.**

**Can establish a
communication plan to keep
them informed**

—

**Identify problems and conflicts
early in order to avoid them.**

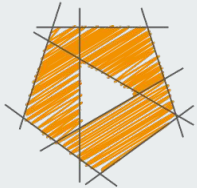






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Architecture Patterns

**“an idea that has been useful
in one practical context and
will probably be useful in
others” - M. Fowler**

—

**TOGAF has the concept of
building blocks: what you use**

**Architecture patterns is: when,
why and how you use**

**Do not currently play a big role
in the TOGAF Standard**

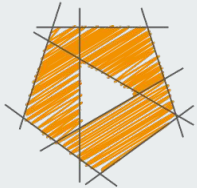
Once you have developed a number of building blocks, you may want to establish patterns for them.





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Gap Analysis

Key step in the BDAT Phases and Migration Planning (Phases B-E)

Baseline architecture

Target architecture

**Gaps are anything added,
changed or intentionally
omitted**

—

**Will also identify things
accidentally omitted**

**It should be clear and easy to
explain why things were added
or removed**

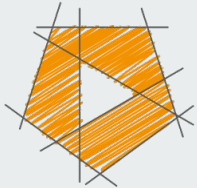
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Migration Planning

**TOGAF contains a number of
tools for migration planning**

—



Migration Planning, Phase F

- Implementation Factor Assessment & Deduction Matrix
- Consolidated Gaps, Solutions, & Dependencies Matrix
- Architecture Definition Increments Table
- Transition Architecture State Evolution Table
- Business Value Assessment Technique



Implementation Factor Catalog

Factors typically include:

- Risks
- Issues
- Assumptions
- Dependencies
- Actions
- Impacts

Implementation Factor Catalog

Factor	Description	Deduction
<Name of Factor>	<Description of Factor>	<Impact on Migration Plan>
Change in Technology	Shut down the message centers, saving 700 personnel, and have them replaced by email.	• Need for personnel training, re-assignment • Email has major personnel savings and should be given priority
Consolidation of Services		
Introduction of New Customer Service		



Consolidated Gaps, Solutions, & Dependencies Matrix

Gaps identified during phase B, C and D.

This matrix can be used as a planning tool when creating work packages.

Work packages identified during phase E and F.



Architecture Definition Increments Table

Architecture Definition Increments table allows the architect to plan a series of Transition Architectures

Lists the projects and then assigning their incremental deliverables across the Transition Architectures



Transition Architecture State Evolution Table

Show the proposed state of the architectures at various levels

At a glance, anyone can see that after Phase 1 of implementation, the new services that will be available even if they are not fully transitioned to the final target architecture yet

Can also use a color coding matrix, with green, yellow and red showing the state of various services after each transition architecture is implemented



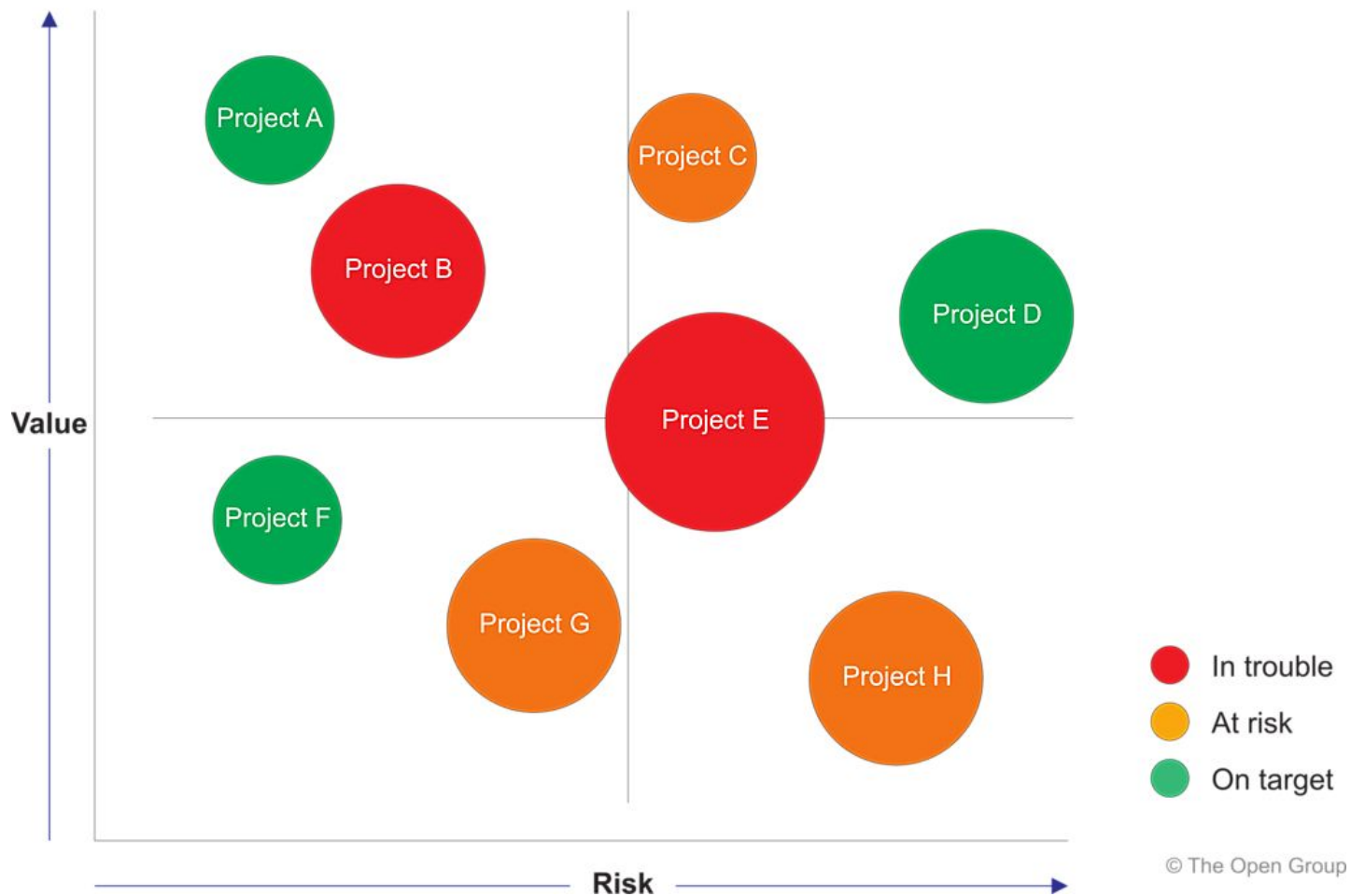
Business Value Assessment Technique

A technique to assess business value based on risk versus value

Another “at-a-glance” way to see the status of the overall implementation project on the overall architecture implementation value

Some projects that are at risk might be low value, which is better than high value projects being at risk

(Project size indicated by size of circle.)

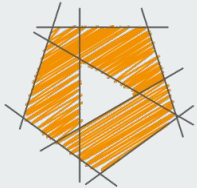






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Interoperability Requirements

**“the ability to share
information and services”**

Business Interoperability: how business teams work together

Information Interoperability: how data is shared

**Technical Interoperability: how
technical services connect to
one another**

—

**Each of the architecture
definition phases has elements
of interoperability.**

—

**Architects think about how
what they do interacts with the
“outside world” during the
process**

—

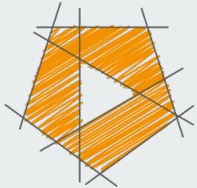


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Business Transformation Readiness Assessment (BTRA)

“evaluating and quantifying an organization's readiness to undergo change”

—

**No point creating architecture
if it'll be ignored.**

Understanding how to get your organization to change is a key to success.



Recommended Activities for BRTA

- Determine the readiness factors that will impact the organization
- Present the readiness factors using maturity models
- Assess the readiness factors, including determination of readiness factor ratings
- Assess the risks for each readiness factor and identify improvement actions to mitigate the risk
- Work these actions into Phase E and F Implementation and Migration Plan



Example Factors

- Vision
- Desire, Willingness, Resolve
- Need
- Business Case
- Funding
- Leadership, Sponsor, Champion
- Governance
- Accountability
- IT Capacity to Execute
- Enterprise Capacity to Execute



Example Factor: Need

How strong is the business need for enterprise architecture?

Does every part of the business recognize this as being required?

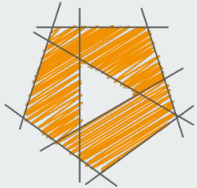
How committed is the senior leadership to this?





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Risk Management

**What are the risks of migrating
to the target architecture?**

—

Identify them, and track them

**TOGAF has the concept of
Initial Level of Risk and
Residual Level of Risk**

—

**Initial Level of Risk is the risk
that exists if you do nothing to
mitigate them**

—

Residual Level of Risk is the risk that remains after you have mitigated them

Example: Client acceptance risk

How do you measure it?

How do you mitigate it?

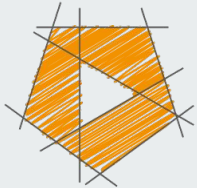
And what would the cost to the business be if it comes true?





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Architecture Alternatives and Trade-Offs

**“a technique to identify
alternative Target
Architectures and perform
trade-offs between the
alternatives”**

—

**Architecture Alternatives:
more than one target
architecture can fulfill the
Architecture Vision**

—

Choosing which architecture to use is a matter of trade-offs



Trade-offs

- Flexibility
- Time and cost
- Length of time to achieve benefits
- Adherence to architecture guidelines
- Solution delivery method (develop, buy)
- Impact on business capabilities
- Risk

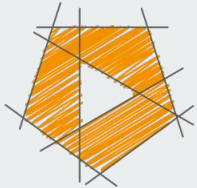


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Architecture Governance

**A system of controls over
architecture**

**A system to ensure compliance
with architecture**

**Effective management of those
systems**

—

**Accountability to the business
for that**

**Governance is layered within
most organizations**

Corporate governance

Technology governance

IT governance

Architecture governance



Principles of Good Governance

- Discipline
- Transparency
- Independence
- Accountability
- Responsibility
- Fairness

Architecture Board

**Representative of all the key
stakeholders in the
architecture**

—

Architecture Contracts

**“joint agreements between
development partners and
sponsors on the deliverables,
quality, and fitness-for-purpose of
an architecture”**

—



Contracts usually cover...

Continuous monitoring for compliance

Adherence to principles, standards and requirements

Identification of risks

Accountability, responsibility and discipline

A formal understanding of the governance organization

Architecture Compliance

- Irrelevant
- Consistent
- Compliant
- Conformant
- Fully conformant
- Non-conformant

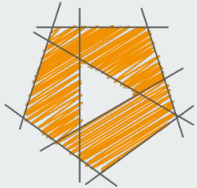
Compliance Reviews





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**Thank You and
Congratulations!**



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